

# Graduation Project Report

**To Identify a Revenue Model for Gravity Gardens to Ensure Financial Stability.**

**Major – International Agri-Business and Trade.**

**Prepared for: Gravity Gardens**

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**Submitted to: Van Hall Larenstein University of Applied Sciences.**

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| <b>Abbreviation</b> | <b>Full form</b>                                                      |
|---------------------|-----------------------------------------------------------------------|
| R&D                 | Research and Development                                              |
| ROI                 | Return on Investment                                                  |
| B2B                 | Business-to-Business                                                  |
| IP                  | Intellectual property                                                 |
| EU                  | European Union                                                        |
| DaaS                | Data as a Service                                                     |
| SaaS                | Software as a service                                                 |
| KPI                 | Key performance indicator                                             |
| NDA                 | Non-disclosure agreement                                              |
| ISF                 | International Seed Federation                                         |
| WBSO                | Wet Bevordering Speur- en Ontwikkelingswerk (Dutch R&D Tax Incentive) |
| GDP                 | Gross Domestic Product                                                |
| VC                  | Venture Capitalist                                                    |

## **Executive Summary**

This graduation thesis explores the development of a revenue model for Gravity Gardens, an innovative Dutch Agri-tech startup specializing in seed priming and molecular farming technologies. The company's flagship innovation, the Hyper Gravity Machine, offers scalable, modular solutions for seed treatment and plant-based compound production. While the technology holds strong scientific promise, the absence of a structured revenue model presents a key barrier to commercial success.

The objective of this research is to identify and recommend a viable revenue model that ensures financial stability for the company. A mixed-methods approach was applied, combining thematic interviews with industry experts and strategic tools such as the ISADA framework and revenue model canvas. The analysis identified EU-based seed companies as the most promising initial customer segment, due to their R&D focus, structured procurement, and openness to innovation.

Customer needs centered on cost-effective, reliable, and non-competitive solutions that integrate smoothly into existing operations. Based on these findings, a hybrid revenue model—combining one-time equipment sales with subscription-based services and performance-linked pricing—is proposed. This approach balances financial predictability with scalability and aligns closely with customer expectations.

The study also reflects on sustainability, showing how Gravity Gardens contributes to multiple UN Sustainable Development Goals (SDGs) through efficient resource use, responsible production, and inclusive innovation. Key recommendations include targeting seed companies, emphasizing value-based pricing, leveraging grants, and maintaining core capabilities in-house. While the research is exploratory and limited by scope and technology maturity, it offers a practical foundation for strategic decision-making.

## A. Introduction

Gravity Gardens, founded in 2024 in the Netherlands, is an innovative startup that designs and develops advanced agricultural and biotechnological machines. The company's core technology centres around the manipulation of gravity within controlled environments, enabling significant improvements in plant growth, seed performance, and compound production.

The Hyper Gravity Machine is the main innovation developed by Gravity Gardens. It is a rotating, vertical structure that looks similar to a vertical farm, with circular trays placed around a central axis. These trays spin at controlled speeds to create a hyper gravity effect, which activates important biological responses in seeds and plants—such as faster germination, better uniformity, and stronger resistance to stress. The machine can be used for various purposes, including seed priming to improve seed quality before planting, molecular farming to grow plants that produce valuable compounds like medicines, and vertical farming to grow crops in stacked layers within small spaces. Its modular and compact design allows it to be scaled up or down depending on the application, and it can be easily integrated into different types of agricultural and biotechnological operations. This makes it a useful tool for sectors such as agriculture, pharmaceuticals, and sustainable biomanufacturing.

Currently, the company employs a dynamic team of 15 young professionals from fields such as plant sciences, embedded systems, mechanical and electrical engineering, computer science, and bioresource technology. With support from renowned institutions including ESA (European Space Agency), SBIC, HAN Innovation Centre, Wageningen University & Research, and Radboud University's Mercator Launch, Gravity Gardens is positioned at the intersection of cutting-edge science and practical innovation. The company is also proudly registered under the Wageningen Starhub and supported by INSPIRO engineering consultancy.



Figure 1- ESA logo



Figure 2- SBIC logo



Figure 3- ML logo



Figure 4- WUR logo

Table 1- Grants and subsidies

| Grants                             | Amount (euros) | Status   |
|------------------------------------|----------------|----------|
| ESA (European space agency grant ) | 60000          | Accepted |
| MIT startup grant.                 | 20000          | Accepted |
| STRATUP VOUCHER                    | 2500           | Accepted |

## B. Problem statement

While Gravity Gardens has developed a cutting-edge solution with multiple applications in agriculture and molecular farming, the economic and commercial aspects have received comparatively less attention. In particular, the lack of a clearly defined revenue model poses a risk to the company's financial sustainability. Without a structured approach to monetization, the startup may struggle to scale operations, attract investment, or compete effectively in the

market. This thesis addresses this gap by focusing on identifying viable revenue model that can support Gravity Gardens' commercialization efforts.

### **C. Research Objective**

To develop a revenue model, which ensures sustainable growth and strengthens profitability for Gravity Gardens.

### **D. Literature Review**

#### **1. Challenges and Risks for Startups**

Despite the abundance of opportunities in today's innovation-driven economy, startups face a daunting failure rate of around 90% globally (Kizilkan, 2024). The most common reasons behind these failures include lack of capital (38%), insufficient market demand (35%), and intense competition (20%) (Kizilkan, 2024). Lack of capital can quickly derail operations, especially when entrepreneurs underestimate the funding needed to reach financial sustainability. At the same time, insufficient demand often stems from a failure to validate the product or service with real customers, leading to offerings that don't resonate with the target market. Competitive pressures further intensify these risks, particularly in saturated sectors where startups struggle to differentiate themselves from established players. To improve their chances of success, startups must prioritize strategic financial planning, ensuring they maintain healthy cash flow and set realistic funding goals (Barros, V. R. 2024). Early customer validation—through MVPs (minimum viable products), pilot launches, or direct feedback—can help align the business with market needs before scaling (Ries, E. 2024, April 29). Additionally, differentiation through unique value propositions, innovative features, pricing models, or strong branding is critical to stand out in a crowded marketplace. While these risks cannot be eliminated entirely, understanding them and planning accordingly gives startups a stronger foundation to survive and thrive in the competitive business landscape.

#### **2. Institutional Factors Influencing Startups in the Netherlands**

The Netherlands has established itself as a leading startup hub in Europe, supported by several institutional factors that drive entrepreneurial activity. These factors include:

##### **2.1. Entrepreneurial Ecosystem and Knowledge Institutions**

The Dutch entrepreneurial ecosystem is shaped by collaboration between businesses, knowledge institutions, and policymakers. Universities and research organizations play a vital role in fostering innovation and entrepreneurship by transferring knowledge to enterprises. However, scaling knowledge-intensive startups remains a challenge, as highlighted by (Henz et al., 2022). Erasmus Centre for Entrepreneurship, 2020 emphasizes the need to improve university-driven entrepreneurial hubs to enhance startup creation and growth (Erasmus Centre for Entrepreneurship, 2020).

##### **2.2. Government Support and Policies**

Generous government initiatives have been implemented to support startups, including the Startup Visa program and tax incentives that attract international talent and investment (Henz et al., 2022). note that these policies contribute to making the Netherlands globally competitive while addressing talent shortages.

##### **2.3. Access to Venture Capital (VC) and Funding**

Although the Netherlands has over 400 VC firms and more than 100 startup accelerators, startups highlight challenges in accessing late-stage funding compared to other ecosystems

like Sweden or Israel. (Chantelleharvey, 2024) points out that increasing domestic capital allocation toward startups could improve scalability rates.

#### 2.4. Sectoral Strengths

The Netherlands is particularly strong in sectors such as health and life sciences, agri-food, semiconductors, and deep tech, identifies these industries as key drivers of the ecosystem's growth due to their concentration of talent and resources. (Startup Genome, 2024).

#### 2.5. Economic Stability and Infrastructure

The stable economy, strategic geographic location, and robust infrastructure of the Netherlands make it an attractive destination for startups. Amsterdam is noted by as one of Europe's fastest-growing startup hubs. (NFIA, n.d.; JourneyBee, n.d.; U.S. International Trade Administration, 2024).

### 3. Introduction to Revenue

Revenue, often referred to as sales or the "top line," represents the total amount of income generated by the sale of goods or services tied to a company's core business operations. It is a fundamental metric that appears at the top of the income statement and serves as a key indicator of financial performance. As defined by Investopedia, *"Revenue is the income generated from normal business operations and includes discounts and deductions for returned merchandise"* (Hayes, 2024). There are two primary types of revenue: operating revenue, which arises from the company's main activities such as product or service sales, and non-operating revenue, which comes from ancillary sources like investments, rental income, or the sale of assets.

Understanding revenue is essential for assessing a business's ability to generate income before accounting for costs and expenses. It also provides insight into the effectiveness of a company's sales and marketing strategies and reflects its overall financial health. Importantly, revenue must be distinguished from profit: while revenue refers to total income, profit (or net income) is the residual amount remaining after all expenses, taxes, and deductions are subtracted.

Beyond its definition as a financial metric, revenue plays a strategic role in a company's business model. Zott (2001) describes a revenue model as a framework that outlines how a business earns income. Building on this, Osterwalder (2004) emphasizes that a revenue model illustrates the logic behind converting delivered value into financial returns, often through multiple revenue streams. Understanding these frameworks is vital for businesses aiming to design scalable and sustainable operations.

### 4. The Concept of Revenue Model

According to TIAS school of business a revenue model framework can be divided into 5 dimensions or as such 5 questions which should be answered 'who pays?', 'what is paid?', 'for what is paid?', 'how is paid?' and 'how much is paid?'. This framework open alternatives to the organizations to think broader and adapt according to situations, instead of following the default pattern of buying selling. This study is intriguing as it explains the fundamentals of revenue model its purpose and the direct benefit or loss. (Meyer, 2023)

### 5. Types of Revenue in Business

Different businesses adopt various revenue models based on their industry landscape and target market. Some of the most used models include recurring revenue (subscription-based), which ensures stable cash flow through periodic payments; affiliate revenue, where

businesses earn commissions on sales generated through referrals; and advertising revenue, where online platforms monetize traffic by selling ad space. Traditional models such as direct sales involve selling products for immediate payment, while leasing and renting allow customers to access high-value equipment without full ownership. Software as a Service (SaaS) has also emerged as a dominant model, where companies provide cloud-based services in exchange for subscription fees (*AgTech Startups: Strategies, Business Models, and Profitable Niches to Consider*, 2024b).

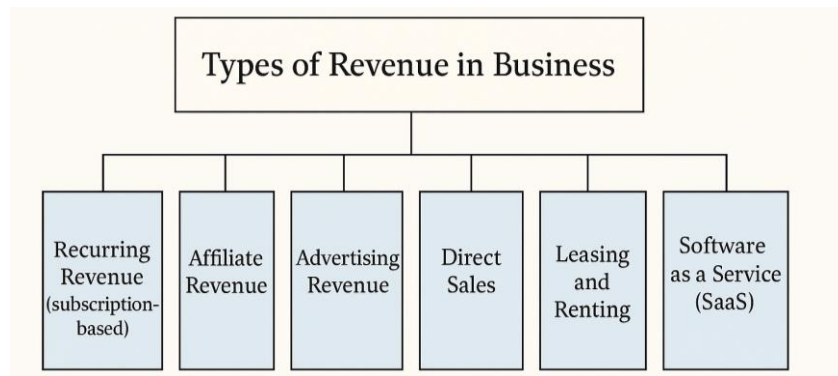


Figure 5 - Types of Revenue model

Organizing for Sustainability- A Guide to Developing New Business Models, by: Jan Jonker, Neils Faber.

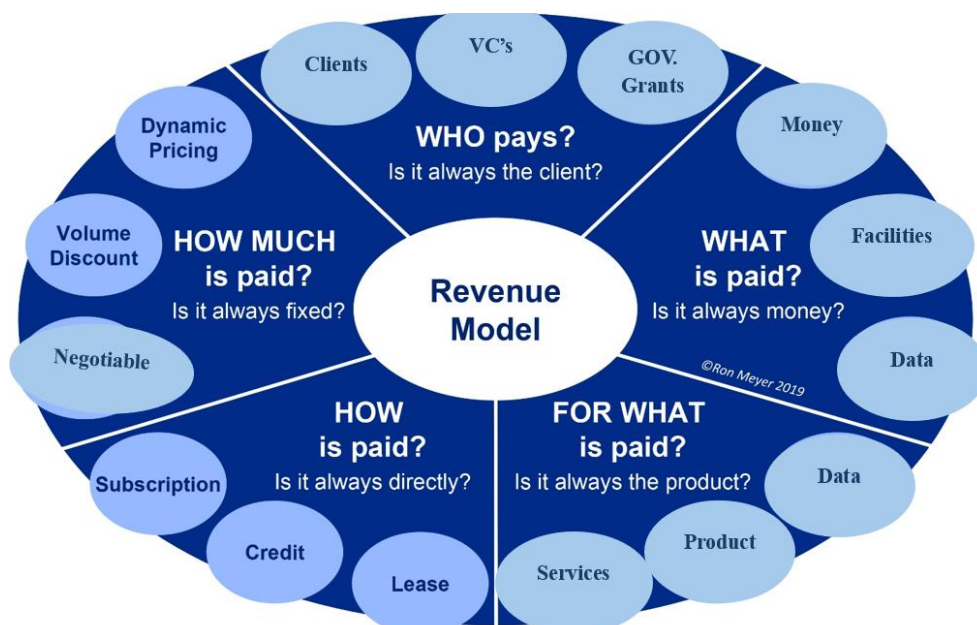


Figure 6- Revenue model framework by Jan Joker

Revenue models, distinct from business models, define how value is generated. Traditional revenue models focus on monetary transactions through ownership transfer or leasing. However, new revenue models include cascade models, where value is stacked in a loop among multiple parties, and hybrid models, where transactions occur without solely relying on money. Hybrid models enable exchanges using alternative transaction mediums, reducing taxation, and fostering local economies. (JONKER, 2021)

Revenue models can be categorized based on ownership transfer or function-based usage, leading to diverse business models such as Product-as-a-Service (PAAS), rental,



performance-based sales, or mixed exchange systems. Hybrid transactions allow businesses to combine financial and non-monetary values, expanding economic participation and sustainability. (JONKER, 2021)

Customer segmentation is the process of dividing a broad and diverse customer base into smaller groups based on specific shared characteristics such as demographics, psychographics, firmographics, and buying behaviour. In Agri-tech, this approach allows businesses to develop tailored marketing strategies, deliver personalized solutions, and respond effectively to global market dynamics, technological advancements, and rising competitive pressure.

However, effective segmentation in the Agri-tech sector presents challenges. Key barriers include the high variability of farming practices, scattered customer locations, and the cost and complexity of accurate data collection and management. These challenges are particularly evident in rural areas, where data infrastructure is limited and operational diversity is high (Chaudhary, 2024). Overcoming these hurdles is essential for delivering value through differentiated business and revenue models.

## 6. Agri-Tech Specific Revenue Models

In the Agri-tech sector, several revenue models are tailored to address the unique needs of farmers and agribusinesses. Data-as-a-Service (DaaS) (Precision Development (PxD), 2025) involves collecting and monetizing agricultural data using AI, IoT, and satellite imagery. Companies like Climate Corporation offer farmers data-driven insights for optimizing crop yields, with pricing based on data granularity and frequency (Wolfert et al. 2017). Equipment-as-a-Service (EaaS) enables farmers to lease expensive equipment like drones and robotic harvesters instead of making large upfront investments. Companies such as XAG provide drone spraying services on a per-acre basis, lowering the cost barrier for small-scale farmers.

Another approach is platform-based revenue models, (Mukherjee, 2024) where digital platforms connect farmers with buyers, suppliers, or financial services, generating revenue through transaction fees or subscriptions (Vlachopoulou et al., 2021). Outcome-based revenue models link payments to measurable results, such as increased yields or reduced input costs. For instance, precision irrigation services may charge farmers based on water savings achieved. Each of these models has distinct advantages; for data-driven startups, DaaS and outcome-based models align well, whereas EaaS is ideal for those dealing with specialized equipment. Platform-based models, on the other hand, benefit businesses facilitating transactions within the agriculture ecosystem (De Franceschi, 2024b).

## 7. Relevant Trends in the Agri-Tech Industry

The Agri-tech industry is rapidly evolving, driven by the urgency to address complex agricultural challenges through technology, sustainability, and innovation. One of the dominant trends is digitalization, where technologies like Artificial Intelligence (AI) and the Internet of Things (IoT) enable real-time monitoring of soil health, weather conditions, and crop growth. These innovations help farmers make data-driven decisions that increase efficiency and reduce resource waste. As a result, the global precision agriculture market is projected to reach \$12.9 billion by 2027, growing at a compound annual growth rate (CAGR) of 12.2% (Gabbio, 2025).

Another significant trend is sustainability, where there is a shift toward eco-friendly agricultural practices. Farmers are reducing their reliance on chemical pesticides and fertilizers, aligning with the rapid growth of the organic food market, which expands at approximately 10% annually (Gabbio, 2025). Regenerative agriculture, which emphasizes soil health, carbon

sequestration, and biodiversity, is growing at a CAGR of 15% through 2027, signalling strong investor and farmer interest in long-term sustainable solutions.

Investment in Agri-tech has surged, with \$16 billion invested in startups in 2024 alone, marking one of the most significant funding years in the industry's history (Gabbio, 2025). Countries like the Netherlands, a recognized hub for Agri-tech innovation, have seen average startup investments of approximately \$2 million. Currently, about 30% of farms globally use precision agriculture technologies—a figure expected to surpass 60% within the next five years (Gabbio, 2025).

A particularly emerging innovation in this field is seed priming using hyper gravity technology. Seed priming, a pre-sowing method to enhance seed germination and vigour, has been traditionally limited by scalability and inconsistent results in field crops. However, hyper gravity-based priming uses centrifugal force to activate metabolic processes in seeds, enhancing uniformity and resilience (Swamy et al., 2021). Early findings suggest that hyper gravity-treated seeds exhibit improved stress tolerance and faster emergence (Yoshihara et al., 2023), making it a strategic advancement for both precision agriculture and molecular farming.

Additional Agri-tech segments are also demonstrating robust growth. Vertical farming, which maximizes land and resource efficiency in urban environments, is expected to reach \$9.7 billion by 2026. Indoor farming, characterized by climate-controlled, pesticide-free environments, is projected to become a \$50 billion industry by 2026. These trends point toward a transformation of traditional agriculture into a highly automated, data-driven, and sustainable ecosystem (AgTech Startups: Strategies, Business Models, and Profitable Niches to Consider, 2024).

## 8. Regulatory and Ethical Considerations

Agri-tech businesses must navigate regulatory and ethical challenges. GDPR compliance is essential when managing farmers' data, requiring robust security measures and transparency in data usage policies. Environmental regulations mandate responsible pesticide, fertilizer, and water management. Additionally, ethical AI use is critical to prevent algorithmic bias and potential job displacement. Companies must also adopt sustainable practices, promoting environmentally friendly alternatives in farming inputs and technology deployment. (GDPR.eu, 2019)

### i. Case Study 1- In Revenue Innovation: Ecosystems, Technology, and Sustainability Cure-4-Hunger – A global solution.



Figure 7 – food domes



Figure 8 – food domes

This is a company from Westlake village, CA, USA. They have developed a product which utilizes micro-gravity to develop the product and have used a very interesting revenue model. This is mass scale selling of product as a packaged deal, which is called the CURE4HUNGER dome. The prices range from \$1,325,000 to \$15,500,000 for 350 to 4,400 units of gravity gardens in a dome as a complete ecosystem quipped with specialized labs for vegetables and plants required for pharmaceuticals (*Cure4Hunger*, n.d.).

ii. Case Study 2 - Business Model Mapping Case Study

This report by Dr. Grey Fox provides an overview of successful companies like Air-bnb and apple and display how these businesses visualize how value flows between partners and stakeholders and ensure that the ecosystem is efficiently structured. This visualization is important because it frames how revenue is directly affected by partners and stakeholders and how companies should leverage this to create a long-lasting business model and a revenue model consecutively. (Fox, 2024)

iii. Case Study 3 - Bayer Annual Reports

Bayer strategically employs various revenue model strategies, including traditional direct sales, alongside cutting-edge service models such as data as a service (DaaS), which offers clients valuable access to comprehensive datasets, analytical tools, and actionable insights pertinent to their operations, and equipment as a service (EaaS), facilitating the utilization of specialized machinery and advanced tools through flexible, subscription-based arrangements rather than requiring an outright purchase. (*Bayer Annual Report*, 2024)

iv. Case Study 4 - John Deere's Innovative Revenue Model

John Deere has pioneered a Solution-as-a-Service (SaaS) model for precision agriculture equipment. Instead of requiring customers to make large upfront investments, the company offers a pay-as-you-go option, providing several advantages: lower initial costs, payments based on actual usage, and continuous technology upgrades over time. This model has significantly expanded John Deere's customer base and ensured steady cash flow through monthly instalment payments rather than relying on large one-time. (Zimmerman, 2024) (*SAAScriptions - Deere's New Path to Technology*, 2024).

9. Market analysis and segmentation

In market analysis and strategic planning, the ISADA model serves as a fundamental framework for assessing the utility of identified market segments. This acronym represents Identifiable, Substantial, Accessible, Differentiable, and Actionable attributes. For a market segment to be considered viable, it must first be identifiable, indicating that its characteristics and constituents can be clearly distinguished and quantified. Second, it must be substantial, possessing sufficient size and purchasing power to justify dedicated marketing investment. Third, the segment requires accessibility, enabling effective reach and service through established distribution and communication channels. Fourth, it should be differentiable, meaning segments exhibit distinct responses to varying marketing mix elements and are conceptually discrete. Finally, a segment must be actionable, allowing for the design and implementation of effective programs to target and serve its members. The application of this model ensures that market segmentation initiatives result in targeted strategies that facilitate efficient resource allocation and the development of tailored approaches for specific customer groups (Kotler & Armstrong, 2021).

## E. Conceptual Model

This conceptual model is designed to illustrate the key components of a revenue model, to illustrate how value is created, captured, and monetized for sustainable financial growth.

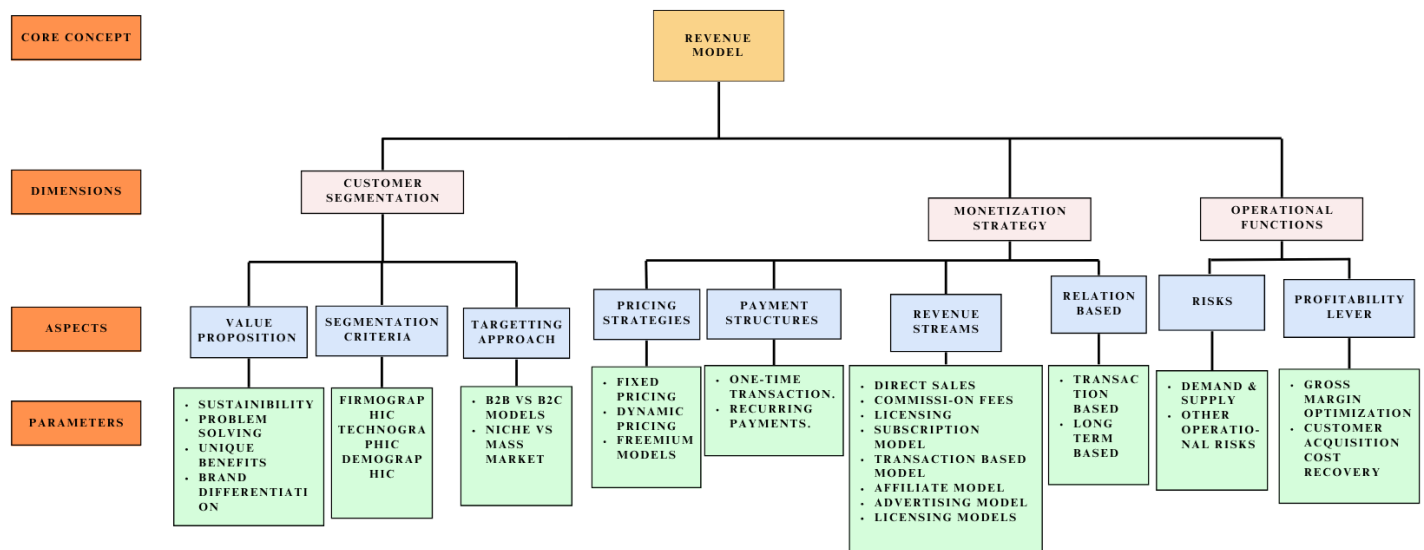


Figure 9- Conceptual model.

## F. Main Research Question

What specific revenue model can Gravity Gardens implement to support a successful launch and drive sustainable growth and long-term profitability?

## G. Sub Research questions.

1. Which is the primary customer segment for Gravity Gardens?
2. What are the needs and preferences of the primary customers of gravity gardens?
3. What are the possible revenue streams for Gravity Gardens chosen customer segment?

## H. The Research Methodology

This chapter describes the research approach used to identify a viable and sustainable revenue model for Gravity Gardens. It includes the research design, data collection methods, data analysis approach, validation techniques, and ethical considerations.

Table 2- The research methodology

| Components                       | RQ1: Primary Customer Segments              | RQ2: Customer Needs & Preferences                              | RQ3: Possible Revenue Streams                       |
|----------------------------------|---------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------|
| <b>Research Approach</b>         | Market Segmentation Analysis                | User-Centred Design & Needs Assessment                         | Revenue Model Exploration                           |
| <b>Primary Data Collection</b>   | No primary data collection                  | In-depth Interviews & with Potential Customers (8 individuals) | Stakeholder & Expert Interviews ( 4 individuals)    |
| <b>Secondary Data Collection</b> | Industry Reports<br>Competitor Demographics | Customer Review Analysis                                       | Competitor Benchmarking<br>Pricing Strategy Reports |
| <b>Data Analysis Techniques</b>  | Comparative Analysis                        | Thematic Analysis                                              | Thematic analysis                                   |
| <b>Outcomes</b>                  | A high potential customer segment           | A list of customer needs, pains, gains, and pain relievers     | A list of revenue streams.                          |

### i. Research Design

The research design presented below outlines the structured approach used to develop a suitable revenue model for Gravity Gardens. It begins by identifying the customer segment (RQ1), followed by an analysis of customer needs (RQ2) and potential revenue models (RQ3). The combined outcomes form the basis for the final recommendations.

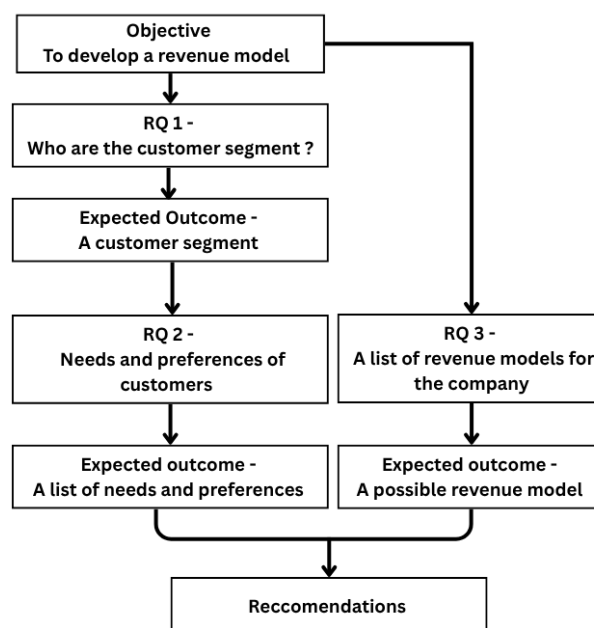


Figure 10- The research design.

## **ii. Data Collection**

### **Primary Data**

Primary data was collected through semi-structured interviews with 8 stakeholders, including representatives from seed companies, seed treatment experts, and technology providers within the Agri-tech and molecular farming sectors. Respondents were selected through purposive sampling, ensuring relevance to the research question. Selection criteria included:

- Experience in seed priming, plant biotechnology, or seed treatment.
- Role in technology adoption or product development.
- Willingness to share insights confidentially.

Interview questions were aligned with themes such as pricing preferences, adoption barriers, and partnership expectations. Interviews lasted between 30–45 minutes and were conducted via Zoom or in-person.

### **Secondary Data**

Secondary data supported the primary findings and included:

- Academic literature on revenue models in Agri-tech.
- Industry reports on seed treatment, vertical farming, and molecular farming trends.
- Company documents and competitor websites for comparative insights.

Sources were triangulated to enhance the reliability and richness of the insights.

## **iii. Data Analysis**

The interviews were recorded (with permission) and transcribed verbatim. The analysis followed a thematic analysis framework inspired by Braun and Clarke (2006). The process involved:

1. Open coding – Initial tags were assigned to meaningful data segments.
2. Axial coding – Similar codes were grouped into broader categories.
3. Theme generation – Overarching themes were developed based on recurring patterns.
4. Interpretation – Themes were linked to research objectives and literature.

NVivo software was not used; instead, coding was done manually in Excel to maintain close familiarity with the data.

#### **iv. Data Validation and Reliability**

Enhance reliability and trustworthiness:

- Member checking was conducted by sharing summaries with a few interviewees for confirmation.
- A transparent coding process was maintained with clear documentation of how codes and themes were developed.

#### **v. Ethical Considerations**

Ethical integrity was maintained throughout:

- All participants gave informed consent prior to the interviews.
- Identities were anonymized to protect confidentiality.
- The data was stored securely and used solely for academic purposes.

No conflict of interest was involved, and the research complied with the ethical guidelines of Van Hall Larenstein University of Applied Sciences.

For this research, the research methodology is framed based on research questions as the number of tools and techniques used for this are vast and the audience for these tools is also diverse, hence this approach is efficient.

- **Expected Outcome**

The expected outcome of this research is to develop a comprehensive and well-justified revenue model that ensures sustainable growth and strengthens profitability for Gravity Gardens. By identifying and evaluating all potential revenue streams—based on market data, customer preferences, and expert insights—the study will recommend the most feasible and strategically aligned model. This outcome will not only support immediate monetization but also provide a robust foundation for long-term financial stability and scalable expansion.

- **Scope of the project**

This project focuses on identifying and evaluating viable revenue models for Gravity Gardens to ensure successful product launch and long-term financial stability. It centres on the commercial potential of the company's innovations in sustainable Agri-tech and molecular farming. The scope is limited to seed and molecular farming applications within the Netherlands, with extended insights from EU farmers, including input from a French farmer. The research emphasizes financial and strategic aspects of revenue generation, rather than technical development, to propose a scalable and customer-aligned model suitable for Gravity Gardens' current phase.

## **I. Results and Findings.**

### **a. Data Analysis for Research Question 1 - Customer Segment identification.**

The aim is to identify which customer group offers the best opportunity for the company to enter the market and grow. Based on internal research and guidance from the commissioner, three customer segments have been selected for comparison in vertical farmers in the Netherlands, seed companies in the European Union, and molecular farming companies in the European Union.

Each of these segments is analysed using the ISADA framework (Kotler & Armstrong, 2021), which looks at five factors: how easy it is to identify the customers, how large the market is, how accessible the customers are, how well the technology fits their needs, and how easy it is to take action in that market. The results of this analysis lead to a recommendation for the most suitable starting point for Gravity Gardens' market entry.

#### **i. Vertical Farmers in the Netherlands: Leadership in Urban Agricultural Innovation**

The vertical farming sector in the Netherlands is one of the most advanced globally, projected to grow from \$79.64 billion in 2024 to \$398.97 billion by 2032, representing a CAGR of 21.86%. The Netherlands holds a unique leadership position as the source of 43% of global Agri-tech patents, making it a prime environment for early adoption of new agricultural technologies. (*Netherlands Vertical Farming Market Size, Trends and Forecast to 2032, 2025*)

Vertical farming operations in the country are known for exceptional resource efficiency, including:

- 98% less water usage than traditional farming.
- 99% less land usage than traditional farming.
- 240x higher yield per square meter through stacked indoor production systems.
- Although use of energy usage in vertical farms is far more than traditional farms.

(Funk & Technology, 2025)

Private and public investments underscore this momentum:

- €20 million invested in companies like Plant Lab, whose systems can feed 100,000 people daily on just two football fields. (Eindhoven, 2020)
- The Oosterwold Urban Farming Cooperative, an 1,800-hectare mixed-use project integrating 15,000 homes with food production. (*Oosterwold: An Example of Peri-urban Agriculture in the Netherlands, 2021.*)

#### **Key Strategic Drivers:**

- Urbanization pressures: Over 93% of the Dutch population lives in cities, increasing the need for hyper-local food systems. (statista.com)
- EU Farm to Fork Strategy compliance: Targets a 50% reduction in pesticide use and 20% in fertilizer use by 2030. (*Farm to Fork Targets - Progress, 2020.*)



- Export scalability: As the world's second-largest agricultural exporter, the Netherlands provides a launchpad for global technology dissemination. (Van Lohuizen/NOOR, 2022)

#### Strategic Relevance

Vertical farming companies require compact, scalable, and precision-controlled machinery to operate effectively in urban and climate-regulated environments. Gravity Gardens' automation and modular technologies align strongly with these demands, offering a strong value proposition to Dutch vertical farmers.

### ii. Seed Companies in the European Union: Stability in a High-Value, Regulated Market

The European seed sector is a mature, innovation-intensive industry valued at €8 billion annually, encompassing 7,000+ propagators and over 200 established seed firms. Prominent players like Bayer Crop Science and Rijk Zwaan account for over 40% of global seed exports, reflecting the sector's global reach and significance. (Marten vd Berg, n.d.)

#### Key Industry Characteristics:

- High R&D investment: Companies reinvest 15–30% of turnover into research and innovation. (Rijk Zwaan, 2024)
- Technological trends:
  - Biological treatments: One important trend is the use of biological treatments on seeds. This means using natural substances like helpful bacteria, fungi, or plant extracts instead of chemical products. These treatments protect the seeds from pests and diseases and help them grow better in a more natural and eco-friendly way.
  - Set to cover 100% of EU-certified organic seeds by 2030. (European Commission, 2021)
  - Traceability and compliance: Widespread integration of blockchain-based systems for GMO-free certification and transparent supply chains.

#### Commercial Accessibility

- Highly structured procurement through platforms like Euro seeds. (*Euroseeds*, 2025)
- 85% of companies actively source new technologies through industry events and trade shows. (*Euroseeds*, 2025)

#### Strategic Relevance

Seed companies present a clear match for Gravity Gardens' technological offerings, particularly in automated seed processing, coating, and traceability solutions. With stable demand, established R&D pathways, and openness to innovation, this segment offers significant opportunity for scalable, B2B technology adoption.

### iii. Molecular Farming Companies in the EU: Pioneering the Agri-Biotech Frontier

Molecular farming—using plants to produce high-value compounds like pharmaceuticals and enzymes—is emerging as a powerful force in the Agri-biotech sector. Although still relatively small, the EU molecular farming market is projected to grow to €500–700 million by 2030, supported by both economic and regulatory drivers. (Next Move Strategy Consulting, 2024).

- Growth Catalysts:

Cost-effective biomanufacturing: Some companies are now using plants to make valuable products like medicines and enzymes, instead of using more traditional methods like growing animal cells in labs (called mammalian cell cultures). Using plants is often cheaper and more sustainable.

Plant-based production systems offer a cost-effective alternative to traditional pharmaceutical manufacturing methods, such as those using animal cell cultures. According to Eidenberger et al. (2023), these systems can reduce production costs by up to 60%, largely due to lower input requirements, simpler growth environments, and scalability in greenhouses or controlled systems.

In parallel, regulatory developments in the EU are making it easier for these innovations to reach the market. The EU Pharmaceutical Strategy Revision (2025) includes measures to streamline the approval process for novel therapies, including those derived from molecular farming. These reforms focus on faster regulatory assessment, clearer guidance on bioproduction, and harmonized standards across member states. Together, these factors accelerate time-to-market for plant-based compounds, helping Europe stay competitive in the growing biopharmaceutical landscape.

#### Strategic Relevance

Despite its small number of players, molecular farming represents a high-value, high-impact customer segment. Gravity Gardens is well-positioned to support these companies with customizable, modular systems suited to regulated, biotech-oriented agricultural production. Additionally, co-development opportunities make this segment especially strategic for long-term partnerships and product innovation.

Evaluate which customer segment is most suitable for Gravity Gardens to target initially, the ISADA model was applied. This framework helps assess potential markets based on five key criteria: Identifiability, Substantiality, Accessibility, Differentiability, and Actionability. By using ISADA, we can systematically compare different segments and determine where the company's technology is most likely to succeed both commercially and strategically.

## ISADA Market selection criteria

Table 3- ISADA Market selection table

| Criteria                    | Vertical Farmers (Netherlands)                                                                                                                                                                                                                                          | Seed Companies (EU)                                                                                                                                                                             | Molecular Farming Companies (EU)                                                                                                                                                          |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifiable                | Yes                                                                                                                                                                                                                                                                     | Yes                                                                                                                                                                                             | Yes                                                                                                                                                                                       |
| Substantial                 | Medium – ~100+ companies, active in urban Agri-tech hubs and listed in innovation databases. Market growing fast (\$79.64B → \$398.97B by 2032, CAGR 21.86%), but many firms face financial difficulties and bankruptcies despite strong national Agri-tech leadership. | High – 200+ firms and 7,000+ propagators, tracked via Euroseeds and Plantum. Strong market: \$23.59B (2024) → \$50.81B (2033), CAGR 8.9%, with Seed Valley (NL) as a major hub. (Forecast 2024) | Medium - <30 active firms, identified through research spin-offs and EU biotech parks. Growing niche market: €500–700M by 2030, CAGR 10–15%, with key hubs in Wageningen, Ghent, and VTT. |
| Accessible in communication | Medium – Companies are present in national events and Agri-tech networks, but the diversity in scale and business maturity may make outreach challenging.                                                                                                               | High – Well-networked through formal platforms like Euroseeds, ISF Congress, and national seed organizations. Direct access via B2B outreach is common and effective.                           | Medium – Often accessed via research institutions and EU-funded collaborations. May require academic or consortium partnerships to initiate contact.                                      |
| Accessible in distribution  | Medium – Products must be adapted to urban infrastructure and small spaces. Modular delivery is possible, but logistics may be complex due to scale differences.                                                                                                        | High – Centralized processing hubs and established supply chains make delivery and installation of technology or services straightforward.                                                      | Medium – May require bespoke equipment or facility adaptation. Product delivery is feasible but involves additional coordination due to regulatory and GMP standards.                     |
| Differentiable              | High – Need for compact, modular, climate-controlled automation. Focus on sustainability, food security, resource efficiency.                                                                                                                                           | High – Specialized machinery for seed treatment, coatings, biologicals. High focus on traceability, compliance, R&D.                                                                            | High – Need for GMP-compliant, precision biotech equipment. Early-stage, innovative, regulatory-driven.                                                                                   |
| Actionable                  | High – Short sales cycles, scalable solutions. Strong government and EU support for urban farming tech adoption.                                                                                                                                                        | High – Structured procurement, R&D-driven culture. Clear product-market fit for precision Agri-machinery and digital solutions.                                                                 | Medium – Requires co-development, pilot integration. Dependent on regulatory and funding cycles.                                                                                          |

From the ISADA framework a conclusion is drawn that seed companies in the EU are the best potential segment for gravity gardens, it is further discussed in the discussion.

## b. Data Analysis for Research Question 2 and 3 – The Thematic Analysis.

To Explore the technological, commercial, and strategic aspects of seed enhancement and molecular farming, 12 semi-structured interviews were conducted with experts from across the agricultural value chain. Participants included seed companies, Agri-tech firms, business experts, molecular farming companies, and a practicing farmer, ensuring diverse insights. Using thematic analysis, six key themes were identified, each reflecting a major area of relevance. Within these, specific codes were developed to capture more detailed expert opinions. The table presents these themes and codes along with their frequency of mention, highlighting the importance of each topic from the interviewees' perspective. Higher frequencies indicate greater relevance, helping prioritize focus areas for developing a market-aligned revenue model.

Table 4 - Themes and codes

| Themes                                    | Codes                                     | Frequency |
|-------------------------------------------|-------------------------------------------|-----------|
| Technological Foundations & Innovation    | Benefits of priming                       | 4         |
|                                           | Gravity manipulation                      | 2         |
|                                           | Strategic innovation pathways             | 4         |
|                                           | Coating Vs Priming                        | 3         |
|                                           | Future outlook and innovation             | 3         |
| Market Landscape and adoption challenges  | Market segmentation                       | 4         |
|                                           | Low adoption in field crops               | 3         |
|                                           | Customer knowledge                        | 4         |
|                                           | Trust and validation barriers             | 4         |
|                                           | Strategic collaborations for market entry | 4         |
| Economic viability and pricing strategy   | Barriers to entry                         | 5         |
|                                           | Original market pricing                   | 3         |
|                                           | Pricing and economics                     | 3         |
|                                           | Revenue model flexibility                 | 4         |
|                                           | Customer willingness to pay               | 3         |
| Strategic growth and commercialization    | Funding and financial strategy            | 1         |
|                                           | Business and commercialization            | 5         |
|                                           | Company strategy                          | 4         |
|                                           | Scalability through financial planning    | 3         |
|                                           | Partnership driven acceleration           | 4         |
| Operational and logistical infrastructure | Long term vision and impact               | 3         |
|                                           | Logistics and processing                  | 4         |
|                                           | Build Vs buy decision                     | 3         |
| Legal and regulatory framework            | Production and supply chain               | 5         |
|                                           | IP and legal protection                   | 5         |
|                                           | Grant and subsidy dependence              | 3         |
|                                           | Role of public institutions               | 3         |

- **Thematic Analysis for Research Question 2**

- i. **Theme 1 – Technological Foundations & Innovation**

In this theme, the interviewees discussed the benefits and importance of seed priming, highlighting its key role in improving germination and uniformity. Many experts emphasized that priming helps seeds germinate faster and more evenly, which is critical for efficient crop production.

### **Benefits of Priming**

Eric J from Centor Europe explained:

*“When you put it through a plant raiser you should prime the seed, because what you get is the seed germinates at the same time very rapidly and very uniformly... If you do not do a priming, you know, the first seed germinates after three days, the last seed germinates after 28 days... By doing a priming you can get a much more uniform result in the end.”*

He further explained that priming is used to break dormancy and achieve uniform maturity in seeds:

*“That’s what you can do with the priming. That’s why as a result you get a uniform and a faster germination because now all the seeds are on the same maturity level... A priming, as I said, can also let’s say overcome dormancy. We had photo dormancy or germination dormancy... you can also overcome with a priming.”*

For example, in lettuce—which is thermally dormant—priming helps seeds germinate over a broader temperature range:

*“Take lettuce for instance. Lettuce is thermal dormant. So, they prime it so that it germinates at a broader temperature range.”*

Several experts stressed that.

at currently, there is no real replacement for priming. Jeroen from De Groot en Slot Seeds stated:

*“We do not foresee replacing priming through classical breeding methods... At present, however, there is no viable technique that can entirely replace priming.”*

Similarly, Eric J added:

*“You will in a certain case, you will never be able to substitute priming, depending on what the requirement is... you can never bring the seed to the same maturity level [with coating].”*

Supporting this view, Patrick from Germaines noted:

*“priming & coating will both play a big role → fundamental for certain plants (small).”*

### **Strategic Innovation Pathways**

Innovation in the seed industry is seen as essential. An expert on the seed business explained:

*“Innovation is essential. The seed industry has become far more data-driven and competitive... If a company isn’t investing in innovation, it’s going to fall behind very quickly. It’s no longer about selling seeds; it’s about offering solutions.”*

However, the industry is traditional and cautious. New products take time to prove themselves under different conditions:

*“It takes time to prove that a new product works consistently.”*

The same expert recommended a long-term approach:

*“Start experiments, validate them, display them and it helps to build relationships. Don’t expect instant acceptance.”*

Regulatory approval can also slow progress:

*“Regulatory approval can also be a bottleneck... It takes time to prove that a new product works consistently... Some of the most promising technologies I’ve seen wouldn’t have survived without early-stage funding.”*

Speaker 1 from Centor Europe advised focusing on a few key use cases:

*“You need to keep a long-term vision. Decide where you want to be and focus your resources there... You’ll always have 5–10 ideas. You must choose carefully.”*

### **Coating vs. Priming**

Rob Pronk from Incotec highlighted that film coating is more commonly used than priming, especially for large-scale field crops:

*“Much more seed is coated than primed, because field crops cover hundreds of millions of acres... Vegetables are planted on a much smaller scale, and priming is a niche sector. So, yes, film coating is used much more than priming.”*

Eric J explained that priming is rarely used for agronomical crops due to the scale, so coatings are enhanced to provide some benefits:

*“Agronomical crops are basically not primed... The bulk volume is so enormous that it’s difficult to do that... So, we look at what can we put in the seed coating to benefit agronomical crops to try and get that same effect.”*

While coatings cannot fully activate the seed's metabolism like priming, they can include biological agents to mimic some priming effects once germination begins:

*“You cannot activate the metabolism of the seed with coating, but you can add certain biologicals or bio-stimulants that trigger similar responses once the seed starts to germinate... That’s something you can do with coating to try and mimic the effect of priming.”*

### **Future Outlook & Innovation**

The market demand for both priming and coating is expected to continue growing. Rob Pronk said:

*“The need for primed vegetable seeds will continue to grow, and we’ll become more efficient in how we provide them.”*

Notes from the Germaines interview reinforced this:

*“Priming & coating will both play a big role – fundamental for certain plants.”*

Jeroen from De Groot en Slot Seeds concluded:

*“Based on our current knowledge and regulations, we do not foresee replacing priming through classical breeding methods... At present, there is no viable technique that can entirely replace priming.”*

Eric J also emphasized:

*“You can never bring the seed to the same maturity level [with coating]... You can’t activate the metabolism like priming does... Coatings can include biologicals, but they only help once germination starts.”*

## ii. Theme 2 – Market Landscape & Adoption Challenges

In this theme, the interviewees discussed how the market for seed priming is segmented, the challenges in adopting priming technology—especially in large-scale crops—and the barriers faced by companies trying to enter this niche. They also highlighted the role of customer knowledge, trust, and strategic collaborations in shaping market dynamics.

### Market Segmentation

Jeroen from *De Groot en Slot Seeds* explained the segmentation between high-value vegetable seeds and large-scale field crops:

*“In the Netherlands, approximately 65% to 70% of seeds we market are primed. This figure can reach 100% for certain varieties.”*

*“In crops like wheat or maize—large-scale, low-cost field crops—the economics of priming are far less favourable compared to high-value vegetable seeds.”*

*“Tomato, onion, lettuce, carrot... small-seeded vegetable varieties are primed. You see the benefit of that.”*

This clearly shows that priming is widely adopted in high-value vegetables, while the economics do not support priming for most field crops.

### Low Adoption in Field Crops

Eric J from *Centor Europe* elaborated on why priming is uncommon for agronomical crops:

*“Agronomical crops are basically not primed... The volume is just too big... the scale of equipment needed becomes too expensive.”*

*“Field crop seeds are just too expensive to dry after priming... it becomes unviable unless the crop is very high value.”*

The large scale and cost constraints make priming impractical for many field crops.

### Customer Knowledge

The interviewees emphasized that farmers often have limited understanding of priming:

Eric J (*Centor Europe*):

*“Very often, the farmer does not know... they are guided by either the chemical company or the seed company for the specific treatment.”*

*“A farmer may have heard of a priming, and he may have seen the effect... but it doesn’t mean that he understands it.”*

*“Seed companies use priming to differentiate... it germinates faster, it’s more uniform.”*

This highlights a knowledge gap where farmers rely heavily on suppliers to select treatments, sometimes without full awareness of the technology’s benefits.

### Trust and Validation Barriers

The seed industry’s traditional nature makes acceptance slow:

An expert on seed business noted:

*“The seed industry is quite traditional, and established players—and customers—are cautious.”*

*“One major barrier is trust. It takes time to prove that a new product works consistently under different conditions.”*

This cautious approach requires new products to build credibility through repeated validation.

## **Strategic Collaborations for Market Entry**

Partnerships with established companies are key for startups:

*The expert explained:*

*“Partnering with established companies can significantly shorten a startup's learning curve.”*

*“Strategic collaboration gives you access to scale, data, and credibility.”*

*“For startups, collaboration isn't just helpful—it's almost necessary.”*

Such collaborations help newcomers overcome barriers and gain market acceptance.

### **Barriers to Entry**

The market is tightly guarded and difficult to penetrate:

*Rob Pronk from Incotec said:*

*“Our priming techniques are kept secret—even from internal employees.”*

*“We can't patent it, so we keep it secret... people who handle chemicals don't know what they're used for.”*

*Jeroen (De Groot en Slot Seeds) added:*

*“Gaining the trust of seed companies is extremely difficult... Seed companies are very cautious about quality... If a batch fails, it can result in massive crop losses.”*

*Eric J (Centor Europe) highlighted the risk of errors:*

*“It's very easy to mess up a batch... and there are thousands of ways to prime incorrectly.”*

*“It would be extremely difficult to enter the market... It's a locked and niche segment.”*

High secrecy, quality demands, and risk make the seed priming market challenging for new entrants.

From the analysis a list of needs and preferences were drawn out.

- **The needs and preferences of customers.**

1. Cost-effective priming solutions for field crops
2. Uniform and faster seed germination
3. Scientifically validated and reliable technology
4. Non-competitive, complementary partnerships
5. Regulatory-compliant and localized processing
6. Flexible and scalable revenue models
7. Alignment with internal R&D strategies
8. High confidentiality and IP protection

- **Thematic Analysis for Research Question 3**

### **iii. Theme 3 – Economic Viability & Pricing Strategy**

This theme explores the pricing landscape for seed priming, the economic considerations for profitability, flexible revenue models, customer willingness to pay, and funding strategies to support innovation.

#### **Original Market Pricing**

The pricing for seed priming varies depending on seed type and scale.

*Rob Pronk from Incotec shared:*



*“Depending on the type of seed, our priming charges are between 20–30 cents and up to 1 euro per 1,000 seeds.”*

Jeroen from De Groot en Slot described their current pricing structure:

*“We currently charge €60 per unit, where one unit comprises 250,000 seeds. This price is subject to annual review.”*

However, Eric J from Centor Europe pointed out the economic challenge in field crops:

*“Priming field crops is not worthwhile... the bulk volume is so enormous... the scale of equipment becomes too expensive.”*

### **Pricing & Economics**

Profit margins are an important factor in economic viability.

Patrick (notes from Germains interview) observed:

*“5% margin is not enough—30% to 60% is required for profitability.”*

Eric J from Centor Europe noted the added value and its acceptance in pricing:

*“The seed just becomes more expensive... that’s value-added. Seed companies know how to calculate.”*

This demonstrates that pricing must reflect the value delivered and align with market expectations for profitability.

### **Revenue Model Flexibility**

Abhijeet Dubey, a revenue model expert, emphasized the need for adaptable business models:

*“A hybrid model is often the most realistic—like offering a core product as a transaction and layering services on top with a subscription.”*

*“Recurring revenue is great for stability, but doesn’t suit every product.”*

*“Transactional sales can work well if the product has a long lifecycle.”*

*“The biggest mistake startups make is choosing a model because it looks good on paper.”*

This highlights the importance of tailoring revenue strategies to the product and market realities.

### **Customer Willingness to Pay**

Pricing should be driven by customer perception and readiness rather than just cost.

Abhijeet Dubey advised:

*“Start with what your customer is willing to pay, not what it costs you to make.”*

*“Price tells a story—about quality, trust, and usefulness.”*

Eric J from Centor Europe added a market perspective:

*“Primed seeds are sold at a higher price... seed companies use it as a differentiator to compete.”*

Understanding customer value perception is key to effective pricing.

### **Funding & Financial Strategy**

Funding plays a crucial role in supporting research and development in this sector.

Sue McKay from Nutrileads explained their approach:

*“We work with research partners and apply for grants to cover costs... Every 30–60 days we submit applications for funding.”*

This illustrates how consistent funding efforts are necessary to sustain innovation and development.

#### **iv. Theme 4 – Strategic Growth & Commercialization**

This theme covers how companies position themselves in the market, their core strategies to avoid conflicts, scalability through financial planning, the importance of partnerships, and maintaining a long-term vision for sustainable impact.

##### **Business & Commercialization**

Companies operate in various roles within the seed and ingredient supply ecosystem, focusing on their strengths without competing directly with their customers.

*Speaker 1 from Nutrileads described their role:*

*“Nutrileads operates as a B2B ingredient supply company. We don’t sell to consumers directly—our goal is to provide high-quality, scientifically validated ingredients to other businesses who then use them in their own products.”*

*Rob Pronk from Incotec explained:*

*“Incotec is a service-based company. We don’t sell seeds ourselves; instead, we specialize in seed treatment services like priming and coating. Our customers are seed companies, and they handle the actual seed sales. We stay focused on helping them improve their product—without becoming their competition.”*

*Similarly, a Germaines representative said:*

*“Germaines is a service provider. We don’t create or market finished products—we focus on adding value to our customers’ seeds through various treatments that enhance performance. That’s our niche.”*

*Jeroen from De Groot en Slot highlighted their collaborative approach:*

*“Our company is also a service provider. We concentrate on specific crops—alliums—and work closely with partners like Bejo Seeds. They handle the priming and processing, while we focus on breeding and commercial distribution.”*

*Eric J from Centor Europe positioned their company as a hardware provider:*

*“We’re a machine manufacturing company. Our role is to provide the hardware and expertise to help seed producers process their seeds efficiently. We don’t produce or treat the seeds ourselves—we support the companies that do.”*

##### **Company Strategy**

Maintaining clear boundaries and building trust are key to long-term business success.

*Rob Pronk from Incotec noted:*

*“Incotec avoids seed production to avoid conflict with clients. Our customers sell seeds, so if we started doing the same, we’d become their competitor. That’s a line we don’t cross—we want to support them, not compete with them.”*

*The Germaines representative shared their approach to early collaboration:*

*“We don’t charge upfront consultancy fees, especially in the early stages. Instead, we try to build trust through collaboration. If things go well, we know we’ll earn their long-term business through product sales, not short-term consulting charges.”*

Eric J from Centor Europe emphasized integrity:

*“For us, integrity and honesty are non-negotiable. In this industry, your reputation is everything. We’d rather tell a potential customer ‘No’ than make a promise we can’t keep. That’s what builds long-term trust.”*

He also explained their focus on long-term partnerships:

*“We’re not in this for one-off consultancy jobs. We’re in it for the long haul—product partnerships that generate value over time. That’s why we focus on selling long-term solutions, not one-time services.”*

Scalability through Financial Planning

Effective financial planning and realistic forecasting are essential for sustainable growth.

Abhijeet Dubey, a revenue model expert, advised:

*“One of the first questions I ask any startup is: can your revenue model grow with the company, or will you have to rebuild it at every stage? Also, startups tend to underestimate fixed costs—things like salaries, compliance, and infrastructure. And when it comes to R&D, look for alternatives—grants, partnerships, shared facilities—whatever helps you stretch your budget without slowing down.”*

He added:

*“Fixed costs will creep up faster than you think. I always advise early-stage founders to build two forecasts: an optimistic one and a conservative one. And manage the company using the conservative one.”*

Speaker 1 from Nutrileads acknowledged the value of grants:

*“Grants really gave us the breathing room we needed in the early days. We could focus on developing the science behind our product without the immediate pressure to generate revenue. That freedom made a big difference.”*

Partnership-Driven Acceleration

Strategic partnerships can provide access to resources, expertise, and markets that accelerate growth.

An expert on seed business noted:

*“If the alignment is right, some partners might offer field trials or even give you access to their R&D setups. Those kinds of partnerships can save you years of effort.”*

They continued:

*“Established companies come with distribution networks, existing credibility, and market access. When you’re a startup, that kind of support can open doors you wouldn’t reach on your own.”*

However, they warned:

*“But partnerships aren’t handouts. You need to bring something of value to the table—whether it’s a novel technology, a strong team, or early results that show potential.”*

*“When you collaborate with the right partner, you can dramatically shorten the learning curve. You get to tap into their expertise and skip a lot of the trial and error.”*

Long-term Vision & Impact

Sustainability and scalability remain core drivers behind company strategies.

Speaker 1 from Nutrileads stated:

*“Our entire approach is about making technology more efficient—not just for today, but for the long run. We want our innovations to scale sustainably.”*

#### **v. Theme 5 – Operational & Logistical Infrastructure**

##### **Code 1: Logistics & Processing**

The Germaines representative emphasized the importance of quality and location in seed treatment:

*“Centralized treatment ensures quality control. That way, we know exactly what we’re delivering and can stand by our product.”*

They further explained the benefits of localizing technology:

*“Localized tech implementation improves logistics. We work directly at the seed production site, so there’s no need to ship seeds across borders for treatment. That saves time and money.”*

Highlighting the critical nature of timing in seed processing, they added:

*“Compressed timelines are crucial for modern seed processing. Farmers and seed companies expect fast turnaround so they can hit the market on time.”*

Finally, Germaines noted the operational pressure:

*“Our processors are expected to deliver ready-to-sell seed quickly. We can’t afford delays because the planting window is short.”*

##### **Code 5: Build vs. Buy Decisions**

Abhijeet Dubey, the revenue model expert, provided insight into strategic decisions around operational infrastructure:

*“Modelling helps quantify the real cost of outsourcing versus developing in-house. It’s not just about today’s price tag—it’s about long-term sustainability and scalability.”*

He cautioned on hidden costs of outsourcing:

*“Outsourcing looks cheaper on paper, but hidden costs can creep in—quality control, logistics, and time delays. A good model helps reveal those.”*

Dubey also pointed out the benefits of internal capacity:

*“Building in-house might seem expensive at first, but it offers long-term savings and more control over your intellectual property.”*

##### **Code 6: Production & Supply Chain**

A Nutrileads representative described sourcing raw materials:

*“Carrot pomace sourcing is done by external agents. We work with farmers and juicing facilities to secure the raw material we need.”*

Germaines clarified their role in the supply chain:

*“Seeds are brought in from the companies, primed, and then sent back to them. We don’t produce seeds ourselves—we just prime and coat them.”*

Jeroen from De Groot en Slot detailed the global nature of their supply chain:

*“Seeds are received from all over the world, then sent back after priming. That way, our customers get the treatment they need, and we maintain high-quality standards.”*

He summarized:

*“Seeds are received primed and sent back. It’s a simple but essential part of the supply chain. We handle the treatment, and our customers take care of the rest.”*

## **vi. Theme 6 – Legal & Regulatory Framework**

### **Code 1: IP & Legal Protection**

Rob Pronk from Incotec shared insights on protecting proprietary technology:

*“Patents vs. trade secrets is a choice you need to make. Our priming process is a secret. We prefer trade secrets because patents can risk leaking of the information.”*

He further emphasized internal confidentiality:

*“Even our employees are unaware of our priming process. That’s how we protect our competitive advantage.”*

An expert added a nuance on patentability for Gravity Gardens:

*“What Gravity Gardens can patent is the specific combination of time and intensity—because you cannot patent the overall use of gravity.”*

### **Code 2: Grant & Subsidy Dependence**

Abhijeet Dubey, the revenue model expert, highlighted the importance of grants for startups:

*“Grants give startups the breathing room to focus on R&D without immediately worrying about revenue.”*

He strongly encouraged pursuing funding opportunities:

*“Grants and subsidies are gold—if you qualify, go for them first.”*

Dubey shared a broader industry perspective:

*“Some of the most promising technologies I’ve seen wouldn’t have survived without early-stage funding.”*

### **Code 3: Role of Public Institutions**

Dubey also spoke about the wider support ecosystem:

*“Governments and grant agencies de-risk innovation and help with mentoring, networking, and validation.”*

He repeated the key benefit of public funding:

*“Public funding can de-risk innovation.”*

And noted the impact of regional initiatives:

*“EU programs or innovation hubs have helped many companies.”*

- Here are the pains and gains identified from thematic analysis and a list of possible revenue models

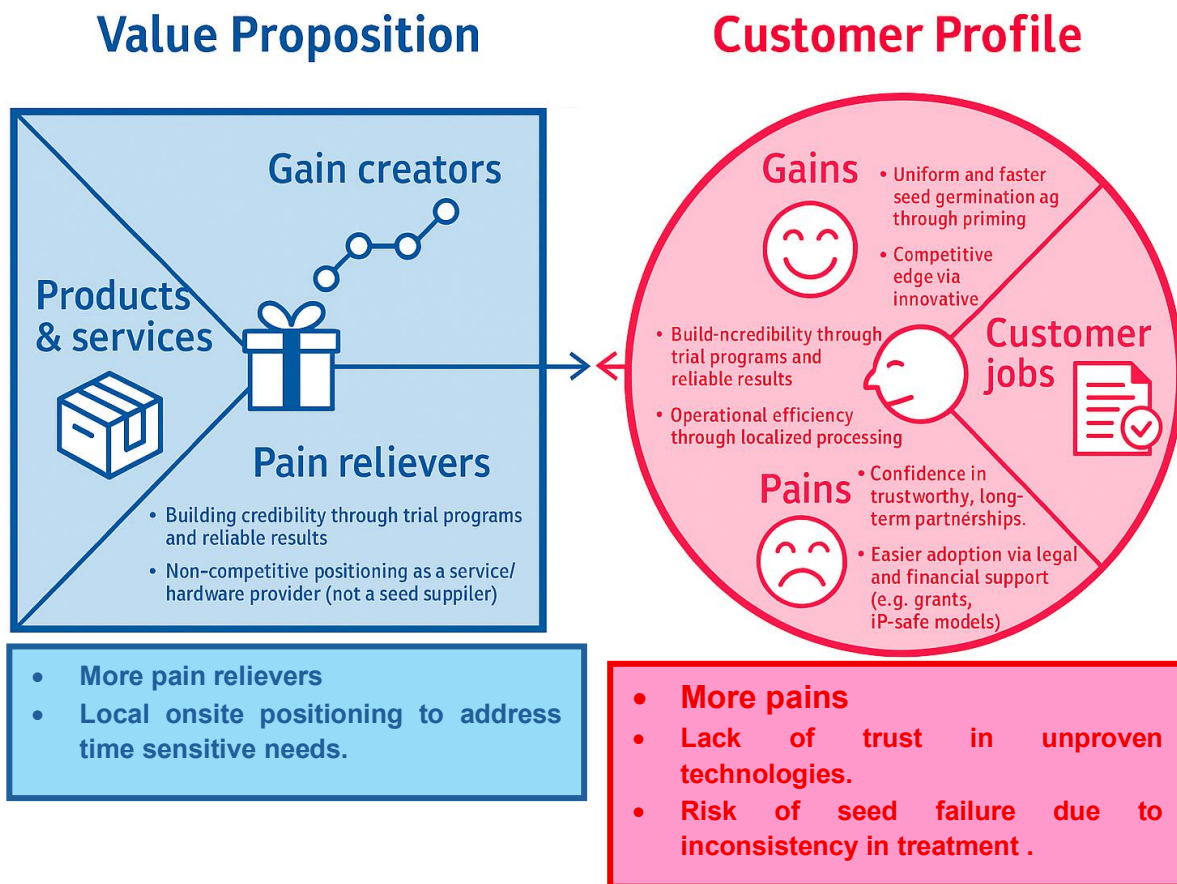


Figure 11- Pains, Gains, and pain relievers

- **The list of possible revenue models.**
  1. Transactional sales
  2. Subscription-based models
  3. Hybrid revenue models
  4. Performance-based models
  5. Licensing models
  6. Freemium/trust-based entry
  7. Grant-funded collaborations

## **J. Discussion**

This chapter discusses the main findings from the results and presents how they answer the three sub-research questions. The goal is to connect what was discovered in the interviews with strategic insights that can help Gravity Gardens make informed business decisions. Where necessary, expert opinions are also included to enrich the interpretation.

### **RQ1. Who are the primary customer segments for Gravity Gardens?**

From Gravity Gardens' strategic standpoint, EU-based seed companies represent the most aligned and high-potential customer segment — especially when considering the company's startup status, innovation model, and cost-sensitive value proposition. While vertical farms and molecular farming companies were considered, seed companies offer a unique fit that supports both early traction and long-term scaling.

One of the most compelling insights from interviews with industry stakeholders is the widespread recognition of priming benefits for field crops — such as improved germination, uniform emergence, and resilience against stress. However, both expert opinion and literature reflect a critical bottleneck: cost. The high price of existing priming technologies has made it economically unfeasible for seed companies to apply these treatments to high-volume, lower-margin crops like cereals, maize, or legumes. Despite strong agronomic evidence supporting the effectiveness of priming, this price-performance gap has discouraged mainstream adoption — creating a clear market opportunity.

This is where Gravity Gardens' USP (unique selling proposition) becomes particularly valuable. By developing low-cost, modular priming systems that can be customized to specific crop requirements, Gravity Gardens directly addresses the economic barrier holding back widespread use. The technology is not just innovative — it's financially accessible, which is crucial for seed companies working with field crops on tight margins. This problem-solution fit positions Gravity Gardens as not just another tech provider, but a partner solving a long-standing commercial challenge in the seed sector.

Moreover, seed companies operate in a structured, research-driven environment. They invest heavily in R&D — often 15–30% of their revenue — and maintain internal teams focused on improving seed quality and performance. This makes them ideal early adopters for co-developing and validating modern technologies. As a startup, Gravity Gardens needs more than just clients — it needs strategic collaborators that can provide feedback, trust, and credibility to build upon. Partnering with large seed firms offers this opportunity, as well as access to their existing customer networks and trade ecosystems.

In contrast, vertical farms, while potentially interested in priming for high-value leafy greens, lack the financial strength or stability to support hardware partnerships. Their market is still maturing and prone to collapse under operational inefficiencies. Molecular farming companies, while technologically aligned, represent a niche and fragmented market where the application of the machine is less relevant to their value chain at this stage.

Importantly, seed companies also offer lower go-to-market risk. Their established procurement structures — including trade fairs, direct B2B platforms, and long-term contracts — allow for quicker pilot deployment, faster feedback loops, and clearer pathways to scale. They also tend to have greater regulatory familiarity, which complements Gravity Gardens' focus on compliance and traceability in seed treatment processes.

In summary, Gravity Gardens is not only solving a technical challenge but unlocking an underexploited market — the priming of field crops — by offering a low-cost, scalable solution

that seed companies are uniquely positioned to adopt. Their financial resources, innovation mindset, and structured business operations make them the optimal starting segment for Gravity Gardens, providing the best conditions for validating, scaling, and expanding this novel agritech solution.

## **RQ 2. What are the needs and preferences of the primary customers of Gravity Gardens?**

For Research Question 2, which focuses on understanding the needs and preferences of Gravity Gardens' primary customers, this part of the discussion looks specifically at customer pains and gains. These insights are important because they help explain how Gravity Gardens can improve its strategy and better meet the expectations of seed companies. One of the main problems mentioned in the interviews and supported by literature is the substantial risk of seed failure due to mistakes during priming. Even small errors can damage a whole batch of seeds, especially in field crops that need to be planted quickly and on time. Because of this, Gravity Gardens must ensure its technology is dependable, easy to use, and reduces the chance of human error.

Another major issue is that many seed companies are hesitant to adopt new technologies without clear proof of performance. Since the seed industry is traditional and careful, companies are unlikely to take risks unless the product has been evaluated in real-world conditions. This means Gravity Gardens should focus on building trust through pilot trials, validation studies, and long-term partnerships. Also, many seed companies prefer to collaborate with partners who do not compete with them directly. There is a concern that a technology provider might later enter the seed market. To prevent this, Gravity Gardens must clearly present itself as a service and hardware provider only—not a seed company.

On the positive side, seed companies are looking for improvements in seed germination. They want faster, more uniform, and stress-resistant seeds, which is exactly what Gravity Gardens' hyper gravity priming technology can deliver. Research shows that this method activates seed metabolism more effectively, leading to better performance in the field (Swamy et al., 2021). In addition, companies prefer solutions that are easy to integrate into their current operations. Gravity Gardens' modular design allows for local or on-site installation, which helps avoid delays. Lastly, if the company can provide secure legal agreements and help clients apply for public funding or grants, it can make adoption much easier. Overall, understanding these pains and gains helps Gravity Gardens build a strategy that is both realistic and attractive to customers in a cautious but innovative industry.

## **RQ 3. What Are the Possible Revenue Streams for Gravity Gardens?**

For Research Question 3, the discussion focuses on why certain revenue models—particularly a combination of transactional sales, subscriptions, and performance-based approaches—are the best fit for Gravity Gardens. The goal is not simply to identify which models to use, but to understand why these models suit the company's current position, market context, and long-term growth plans. As a startup entering a cautious and specialized industry like Agri-tech, Gravity Gardens needs a revenue structure that provides stability, builds trust, and supports gradual scaling without overextending resources.

A hybrid revenue model is particularly suitable because it allows the company to meet both short-term financial needs and long-term strategic goals. One-time equipment sales provide upfront revenue that is essential in the early stages of the business. This supports manufacturing, logistics, and initial operations without relying heavily on external funding. However, relying solely on hardware sales would expose the company to inconsistent cash



flow and unpredictable sales cycles—common risks for startups in hardware-focused industries. By combining equipment sales with recurring revenue through subscriptions for software, remote monitoring, or after-sales services, Gravity Gardens creates a financial model that is more balanced and sustainable over time.

These models also reflect the way value is delivered to the customer. Gravity Gardens is not just offering a physical machine—it is providing a performance-based solution that improves seed quality, germination speed, and consistency. This makes value-based and performance-linked pricing models a natural fit. From the company's perspective, these models allow it to align its success with the success of its customers. When clients see improved outcomes, they are more willing to invest—and this reinforces customer loyalty and builds Gravity Gardens' reputation as a high-impact solution provider. Offering pricing based on delivered value also allows the company to capture a fair return for the improvements its technology makes, rather than being limited by traditional cost-plus pricing.

Scalability is another reason these models are a strong fit. As the company grows, its revenue model must support expansion without a sharp rise in costs. Subscription and licensing models, once implemented, allow the company to serve more customers without having to increase infrastructure and staffing at the same rate. This is important because fixed costs such as technical support, compliance, and equipment maintenance can quickly become a burden for a small company. With a scalable model in place, Gravity Gardens can grow gradually, while maintaining control over its costs and service quality.

From a strategic standpoint, offering flexible entry options such as freemium trials or low-cost pilot projects also supports early customer engagement. In a conservative industry, clients often hesitate to invest in untested solutions. Gravity Gardens can use these entry models to build trust and demonstrate value before moving to full-scale commercial agreements. This staged approach fits well with the relationship-driven nature of the seed industry, where long-term collaboration often begins with small, low-risk commitments.

Finally, grant-funded collaborations and non-dilutive public funding are not just financial boosters—they are also strategic tools that reduce pressure on early revenues. For Gravity Gardens, accessing innovation grants or EU subsidies allows the company to continue R&D, run pilot projects, and improve its product without rushing to secure immediate profits. This breathing space is essential for refining technology, building case studies, and proving value—all of which support the company's entry and positioning in the market.

In summary, these revenue models are a strong fit for Gravity Gardens not only because they provide multiple income streams, but because they are aligned with the company's innovation-driven value proposition, limited early-stage resources, and the structure of the industry it is entering. They support financial stability, gradual scaling, stronger customer trust, and long-term competitiveness—making them the most strategic option for the company's growth and sustainability.

### **Sustainability and Alignment with SDGs**

An essential dimension of Gravity Gardens' value proposition lies in its alignment with global sustainability goals. The company's core technology directly contributes to reducing input waste, increasing seed efficiency, and promoting resource-conscious agriculture—addressing SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption and Production). By enabling

precise seed priming and reducing dependency on chemical seed treatments, Gravity Gardens also supports SDG 13 (Climate Action) by promoting low-impact agricultural practices. Additionally, the modular and on-site nature of the Hyper Gravity Machine reduces logistical emissions and supports SDG 9 (Industry, Innovation and Infrastructure) through sustainable industrialization. The use of public grants for R&D and non-competitive B2B models fosters inclusive innovation, contributing to SDG 17 (Partnerships for the Goals). These sustainability elements not only enhance Gravity Gardens' strategic appeal but also position the company as a responsible innovator in the Agri-tech ecosystem, increasing its value proposition to impact-driven investors and partners.

- **Limitations to this research**

While the study offers valuable insights for developing a revenue model for Gravity Gardens, it faces several limitations that affect its scope, generalizability, and depth of analysis.

**Key Limitations:**

1. **Limited sample size and regional focus** – Most interviews were with European stakeholders, which may not reflect global perspectives.
2. **Qualitative data reliance** – Thematic analysis of interviews introduces a degree of subjectivity and limits statistical generalization.
3. **Lack of real-world testing** – Proposed revenue models were not piloted or implemented, so feasibility remains theoretical.
4. **Assumptions in legal and regulatory areas** – Certain IP and grant-related recommendations are based on expert opinion, not formal audits.
5. **Emerging stage of the technology** – The hyper gravity-based priming technology is still early-stage, making some recommendations speculative and dependent on future technical success.

## **K. Conclusion**

This research set out to identify the most promising customer segment, understand customer needs and preferences, and explore viable revenue streams for Gravity Gardens — an early-stage Agri-tech startup offering seed treatment technology. By conducting expert interviews and applying thematic analysis, supported by structured models such as ISADA, this study has produced a strategic roadmap to guide the company's initial market entry and business model design.

### **RQ1. Who are the primary customer segments for Gravity Gardens?**

Using the ISADA framework, three potential customer segments were evaluated: vertical farmers in the Netherlands, EU-based seed companies, and molecular farming companies in the EU. Among these, EU-based seed companies were found to be the most attractive segment for Gravity Gardens. This group offers a stable, innovation-driven environment with established procurement processes, high R&D investment, and clear alignment with Gravity Gardens' value proposition. While vertical farming and molecular farming present exciting opportunities, they currently carry higher risks due to financial instability and market immaturity, respectively.

### **RQ 2. What are the needs and preferences of the primary customers of Gravity Gardens?**

The thematic analysis of expert interviews revealed that customers in the seed sector prioritize product reliability, operational efficiency, and strategic collaboration. They seek partners — not just suppliers — who can offer long-term value without becoming competitors. Customers value technologies that are tailored to their specific crop needs, deliver consistent performance, and integrate smoothly into existing workflows. Financial and regulatory concerns also play a role, with customers expecting technology providers to demonstrate both legal awareness and access to public funding mechanisms. These findings highlight the importance of building trust, offering proven results, and remaining flexible in how products and services are delivered.

### **RQ 3. What Are the Possible Revenue Streams for Gravity Gardens?**

The findings suggest that Gravity Gardens should adopt a hybrid revenue model that combines one-time equipment sales with recurring service-based income such as software subscriptions or analytics support. This model aligns with the financial logic of B2B customers and provides a stable foundation for growth. Value-based pricing is essential, with a focus on demonstrating ROI to justify premium pricing. Strategic decisions around outsourcing, intellectual property, and in-house development must be made carefully to maintain long-term profitability. Furthermore, public grants and EU innovation subsidies are valuable sources of early-stage funding that can help de-risk the business and accelerate time-to-market.

## L. Recommendations

To guide strategic decision-making, this section begins with an adapted version of the revenue model canvas developed by Jan Jonker. This framework helps visualize the structure of a company's revenue model by answering five key questions: who pays, what is paid, for what is paid, how is it paid, and how much is paid. The customized canvas below outlines Gravity Gardens' potential revenue streams in a simplified and strategic format. It identifies seed companies and government grants as the primary sources of income, while payments can include not only money but also R&D support. Customers may pay for a combination of machines, technical support, maintenance, and data services, using methods such as subscriptions, full payments, or hybrid options. The pricing itself is negotiable and depends on customization, with models also allowing for percentage-based payments. This canvas provides a clear foundation for the recommendations that follow, ensuring they are based on a flexible, scalable, and customer-aligned revenue structure.

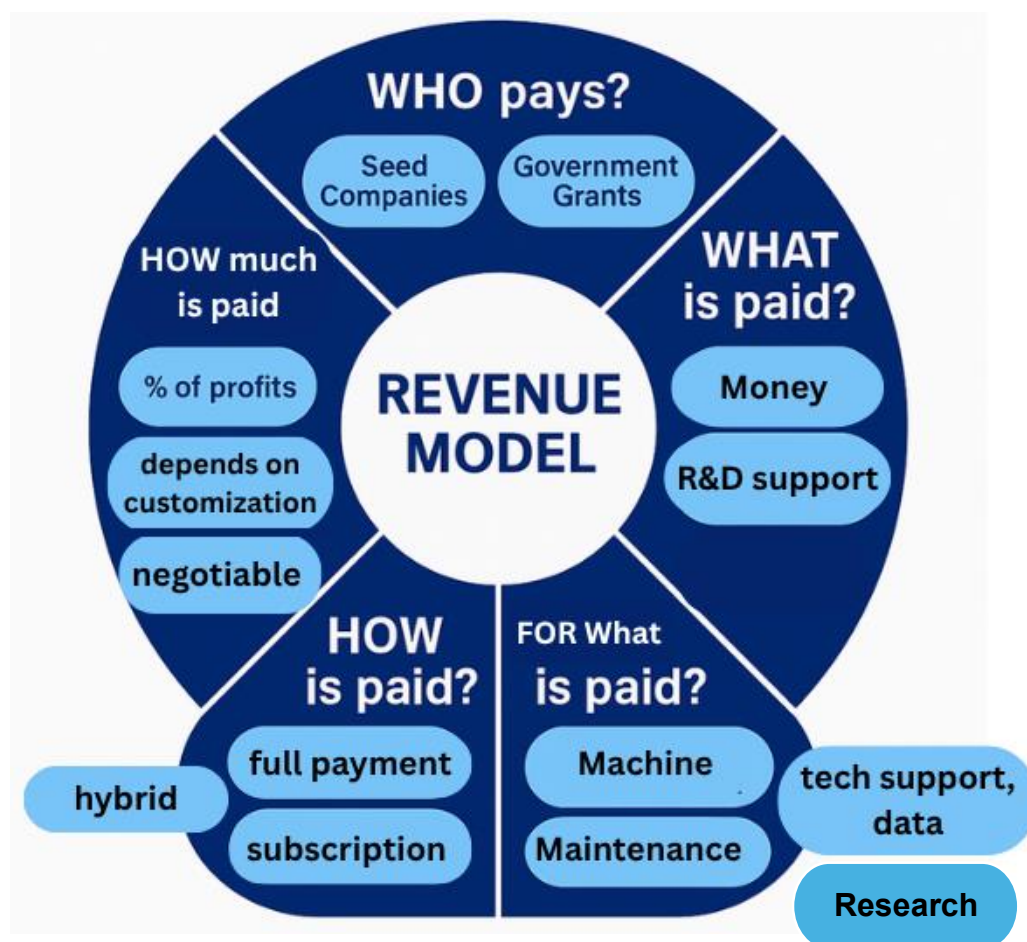


Figure 12 - The final revenue model

In addition to the revenue model canvas, which outlines key elements such as who pays, what is paid, for what, how, and how much, the following are extra strategic recommendations tailored to support Gravity Gardens' successful market entry, scalability, and long-term competitiveness in the Agri-tech sector:

- a. **Prioritize EU-Based Seed Companies as Initial Market Segment**  
Gravity Gardens should focus its market entry on seed companies within the European Union, particularly in the Netherlands (e.g., Seed Valley). This segment is stable, substantial, and innovation-driven, offering a reliable and scalable market with structured

procurement systems. These companies are also actively seeking automation solutions in seed treatment and are familiar with trade show-based procurement.

Why:

Seed companies demonstrate strong alignment with Gravity Gardens' core technology. Targeting them first allows for faster sales cycles and validation of the product in a lower-risk environment.

Practical Step:

Attend ISF World Seed Congress, Euroseeds, and Seed Meets Technology events.

Prioritize outreach to companies like Rijk Zwaan, Incotec, or Bayer.

b. Leverage Value-Based Pricing Aligned with ROI

Gravity Gardens should price its products based on the value they create — such as improved germination rates or regulatory compliance — rather than production cost.

Why:

Seed companies are willing to invest in technologies that offer measurable gains. Value-based pricing also supports premium positioning.

Practical Step:

Create crop-specific ROI calculators (e.g., yield gains per 1,000 seeds) to use in sales discussions.

c. Emphasize Customization and Co-Development

Offer project-based services to clients with specific needs, such as molecular farming startups or niche crops. Position this as a premium, collaborative offering.

Why:

Co-development builds trust and long-term partnerships while allowing Gravity Gardens to evolve with customer needs.

Practical Step:

Introduce paid pilot programs for custom protocols or hardware configurations.

d. Strategic Participation in Sector Trade Events

Proactively participate in industry events where B2B procurement and technology scouting occur.

Why:

Trade fairs are key points for client discovery and relationship-building in the seed industry.

Practical Step:

Display demo units, run live case studies, and collect leads at events like Euroseeds and ISF.

e. Strengthen Position as a Non-Competitive Enabler

Make it clear that Gravity Gardens is not a seed producer but a technology enabler that empowers existing players.

Why:

This avoids industry suspicion and supports strategic partnerships rather than competition.

Practical Step:

State explicitly on all platforms: "We do not produce seeds — we enhance your ability to."

f. Secure Intellectual Property and Maintain Trade Secrets

Protect core priming methods as trade secrets but patent specific time-intensity protocols and mechanical components that are novel.

Why:

This protects competitive advantages and reassures investors without exposing core IP.

Practical Step:

File patents for key configurations and implement NDAs for sensitive internal processes.

g. Apply for EU Innovation Grants to Reduce Capital Risk

Seek funding from Horizon Europe, WBSO, and Dutch regional innovation schemes.

Why:

non-dilutive grants reduce financial pressure and fund R&D without giving up ownership.

Practical Step:

Work with a grant writer to prepare a roadmap for recurring applications across different EU calls.

h. Retain Core Capabilities In-House for Competitive Edge

Keep control over key areas such as priming algorithms, machine design, and software interfaces.

Why:

Protecting these areas ensures innovation continuity and safeguards intellectual property.

Practical Step:

Outsource only non-critical elements like casing production or logistics.

i. Plan for Modular, Localized Infrastructure

Design modular, plug-and-play Hyper Gravity Machines that can be deployed on-site at seed processing centres.

Why:

On-site deployment enhances adoption by reducing logistics, increasing convenience, and improving ROI.

Practical Step:

Develop a compact, containerized version of the system that can be easily shipped and set up anywhere in Europe.

## **M. Suggestions for Future Research**

- To build on this study, future research could:
- Include a quantitative validation of pricing models through customer surveys.
- Explore regional differences in revenue model acceptance across other EU or global seed markets.
- Conduct pilot projects to test the financial impact of the proposed hybrid model in real-world settings.

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## ii. Annexures

### Annex 1 - Interview Introduction Script

Hello, and thank you for agreeing to participate in this interview. My name is Atharva Janjire, and I am a student at Van Hall Larenstein University of Applied Sciences, currently completing my thesis as part of the International Agri-business and Trade program. The purpose of my research is to explore suitable market opportunities and revenue models for an agritech startup called Gravity Gardens, focusing particularly on seed treatment automation and technology adoption in various agricultural sectors. This interview is part of the data collection for my thesis and will take approximately 20 to 30 minutes. It will be semi-structured — I have prepared some key questions, but I also encourage you to share any relevant insights or experiences that may go beyond these questions. Your input is highly valuable and will contribute directly to developing practical recommendations and understanding current trends in the aggrotech industry.

#### ▪ Consent Statement / Ethics Disclaimer

Before we begin, I would like to ensure that you are fully informed and comfortable with your participation. Please note the following:

- Voluntary participation: Your participation in this interview is completely voluntary. You may choose not to answer specific questions or withdraw from the interview at any time without any consequence.
- Confidentiality: All information shared during the interview will be treated with strict confidentiality. Your name, organization, and any identifying information will not be

published. In the report, responses will be anonymized using generic titles (e.g., “Seed Company Manager” or “Expert Interviewee”).

- Data usage: The data collected will be used solely for academic purposes related to my thesis and will not be shared externally or used for commercial gain.
- Recording: With your consent, the interview may be audio recorded to ensure accurate transcription and analysis. These recordings will be securely stored and deleted after the thesis is submitted and evaluated.

## ***Annexures 2 – interviews with individuals from respective field.***

*Table 5 - the interviewee list*

| <b>Interviewee</b> | <b>Company</b>                      | <b>Designation</b>                                     | <b>Contact</b>                                                                           |
|--------------------|-------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------|
| Rob Pronk          | Incotech                            | Global Marketing Manager and Global Regulatory Manager | <a href="mailto:Rob.pronk@incotech.com">Rob.pronk@incotech.com</a>                       |
| Sue McKay          | Nutrileads                          |                                                        | <a href="mailto:sue.mackay@nutrileads.com">sue.mackay@nutrileads.com</a>                 |
| Eric Jukers        | Germaines                           | Representative                                         | <a href="mailto:Eric.jukers@germaines.com">Eric.jukers@germaines.com</a>                 |
| Ravi Kumar         | Formerly NASA, professor in Dharwad | Professor                                              | <a href="mailto:hosamanirr@uasd.in">hosamanirr@uasd.in</a>                               |
| Abhijeet Dubey     | Expert                              | Owner of a business in manufacturing                   | <a href="mailto:abhijeet.dubey@gmail.com">abhijeet.dubey@gmail.com</a>                   |
| Farmer             | A farmer from France                | Bruno Delattre                                         |                                                                                          |
| Klaas Reimpa       | Phytopharma                         | Pharmacist and consultant                              | <a href="mailto:klaasriempa@gmail.com">klaasriempa@gmail.com</a>                         |
| Ab Aarnoudse       | Seed Processing NL                  | Representative                                         | <a href="mailto:Ab.aarnoudse@seedprocessing.nl">Ab.aarnoudse@seedprocessing.nl</a>       |
| Jeroen Kuin        | De Groot n Slot                     | sales manager                                          | <a href="mailto:j.kuin@degrootnslot.nl">j.kuin@degrootnslot.nl</a>                       |
| Umeera Khan        | Private management consultant       | Expert                                                 | <a href="mailto:umerakhan34@gmail.com">umerakhan34@gmail.com</a>                         |
| Paulino Valdes     | Gravity gardens                     | CEO                                                    | <a href="mailto:paulionvaldes@gravity-gardens.com">paulionvaldes@gravity-gardens.com</a> |
| Robin Le Vigouroux | Gravity gardens                     | COO                                                    | <a href="mailto:robinlevigouroux@yahoo.fr">robinlevigouroux@yahoo.fr</a>                 |

## Annex 3 - Thematic analysis

This annex outlines the thematic analysis based on the transcribed interviews conducted for this research. The transcripts were carefully reviewed, and key statements were first broken down into initial codes reflecting core ideas. These codes were then grouped into broader themes relevant to the research objectives, such as adoption barriers, market readiness, and procurement behaviour.

The table below presents each theme, its related codes, supporting quotes, and repetition patterns observed across interviews. This structured process helped ensure consistent interpretation and reliable insights that informed the final recommendations.

| Themes                             | Codes                                     | quotes                                                                                                                                                                                                                      | Repetitions of Codes                                                                        | Repetitions of Codes                                                                                    | Repetitions of Codes                                                        | Repetitions of Codes                                                                                                                |  |  |                 |
|------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--|--|-----------------|
| Technological Foundations & In     | Benefits of priming                       | "Priming improves germination speed and uniformly"                                                                                                                                                                          | Priming used for breaking dormancy and achieving uniform maturity                           | No current replacement for priming (e.g., for dormancy/uniformity)                                      |                                                                             | Priming and coating remain fundamental technologies                                                                                 |  |  |                 |
|                                    | Gravity Manipulation                      | "Hypergravity application can change the results of crop seeds"                                                                                                                                                             | It is a promising field to work on and it is predicted to have amazing benefits.            |                                                                                                         |                                                                             |                                                                                                                                     |  |  |                 |
|                                    | Strategic Innovation Pathways             | "Start experiments, validate them, display them and it helps to build relationships. Don't expect instant acceptance."                                                                                                      | It takes time to prove that a new product works consistently."                              | Innovation is absolutely essential."                                                                    | "Focus on one or two key use cases where your technology can really shine." |                                                                                                                                     |  |  |                 |
| Market Landscape & Adoption Cl     | Coating vs. Priming                       | "Film coating is dominant due to large-scale field crop usage"                                                                                                                                                              | Coatings sometimes preferred in agronomic crops as it is cheap                              | Coatings may be improved to mimic priming benefits                                                      |                                                                             |                                                                                                                                     |  |  | incotec         |
|                                    | Future Outlook & Innovation               | "Market will continue to demand priming and coating"                                                                                                                                                                        | we do not see possibilities of priming being replaced by anything soon                      | Coatings helps till an extent but not efficient as priming                                              |                                                                             |                                                                                                                                     |  |  | germains        |
|                                    | Market Segmentation                       | "Vegetable seeds: priming widely adopted"                                                                                                                                                                                   | Field crops: priming not common due to cost                                                 | High-value crops use priming; low-margin field crops                                                    |                                                                             | Tomato, onion, lettuce, carrot - commonly primed                                                                                    |  |  | de grootn slot  |
|                                    | Low Adoption in Field Crops               | "Maize, wheat - rarely primed"                                                                                                                                                                                              | Priming not feasible for low-value, large-volume                                            | Low ROI for field crop priming                                                                          |                                                                             |                                                                                                                                     |  |  | nutrileads      |
|                                    | Customer Knowledge                        | "Farmers often lack technical understanding of priming"                                                                                                                                                                     | Farmers follow recommendations from seed/chemical companies.                                | Priming used by companies to differentiate seeds: Eg Primed seeds and unprimed seeds                    |                                                                             | Awareness and education efforts are ongoing                                                                                         |  |  | ravi humar      |
| Economic Viability & Pricing Stra  | Trust and Validation Barriers             | "The seed industry is quite traditional, and established players—and customers—are"                                                                                                                                         | "One major barrier is trust."                                                               | "It takes time to prove that a new product works."                                                      |                                                                             | Farmers often lack technical understanding of priming."                                                                             |  |  | centor          |
|                                    | Strategic Collaborations for Market Entry | "Partnering with established companies can significantly shorten a startup's learning curve."                                                                                                                               | "You don't want to do it all alone—especially not in this industry."                        | "Strategic collaboration gives you access to scale, data, and credibility."                             |                                                                             | "Collaboration isn't just helpful—it's almost necessary."                                                                           |  |  | farmer          |
|                                    | Barriers to Entry                         | "Priming techniques are trade secrets, not patented so can be copied."                                                                                                                                                      | Internal compartmentalization protects the priming process                                  | Difficult market to enter due to complexity and trust with the customer (eg. Spald) companies           |                                                                             | Seed companies have built years of relationships and trust with their priming companies, making it difficult to switch to new       |  |  | seed processing |
|                                    | original market pricing                   | "we price around 30 cents to a euro"                                                                                                                                                                                        | we charge around 80 euros for a unit consisting of 250000 seeds                             | priming field crops is very expensive                                                                   |                                                                             | High learning curve; risk of seed batch failure, which can result in loss of farmers for the whole crop                             |  |  | expert 1        |
|                                    | Pricing & Economics                       | "Our pricing is based on seed quantity per 1,000 seeds."                                                                                                                                                                    | Profit margins for priming must be 30-60% for profitability                                 | Primed seeds sold at a higher price of almost 50 % or 100%; too.                                        |                                                                             |                                                                                                                                     |  |  | expert 2        |
| Strategic Growth & Commercializ    | Revenue Model Flexibility                 | "A hybrid model is often the most realistic—like offering a core product as a transaction and layering services on top."                                                                                                    | "Recurring revenue is great for stability, but doesn't suit every product."                 | "Transactional sales can work well if the product has a long lifecycle."                                |                                                                             | "The biggest mistake startups make is choosing a model that looks good on paper."                                                   |  |  |                 |
|                                    | Customer Willingness to Pay               | "Start with what your customer is willing to pay, not what it costs you to make."                                                                                                                                           | "Price tells a story—about quality, trust, and usefulness."                                 | "Primed seeds sold at a higher price"                                                                   |                                                                             |                                                                                                                                     |  |  |                 |
|                                    | Funding & Financial Strategy              | "grants like SDR is first initial streams of income for research"                                                                                                                                                           |                                                                                             |                                                                                                         |                                                                             |                                                                                                                                     |  |  |                 |
|                                    | Business & Commercialization              | "nutrileads is a B2B ingredient supply company. Incotec avoids seed production to avoid conflict with clients"                                                                                                              | incotec is a service-based company. Germains avoids upfront consultancy fees to build trust | germains is a service provider company. Integrity and honesty are key to building long-term trust       |                                                                             | de grootn slot is a service provider company.                                                                                       |  |  |                 |
|                                    | Company Strategy                          | "Can the model grow with the company without needing to constantly rebuild it? don't underestimate anything and look for alternatives for some partners might offer field trials or R&D support—if the alignment is right." | "Germains often underestimate fixed costs."                                                 | "Grants give startups the breathing room to focus on R&D."                                              |                                                                             |                                                                                                                                     |  |  |                 |
| Operational & Logistical Infrastru | Partnership-Driven Acceleration           | "Established companies have distribution networks and credibility."                                                                                                                                                         | "You need to bring something of value to the table."                                        | "Partnering with them can significantly shorten the learning curve."                                    |                                                                             |                                                                                                                                     |  |  |                 |
|                                    | Long-term Vision & Impact                 | "to make technology more efficient"                                                                                                                                                                                         | farmer empowerment                                                                          |                                                                                                         |                                                                             |                                                                                                                                     |  |  |                 |
|                                    | Logistics & Processing                    | "Centralized treatment ensures quality control"                                                                                                                                                                             | Localized tech implementation improves logistics                                            | Compressed timelines for modern seed processing                                                         |                                                                             |                                                                                                                                     |  |  |                 |
|                                    | Build vs. Buy Decisions                   | "Modeling helps quantify the real cost of outsourcing versus developing in-house."                                                                                                                                          | "Outsourcing looks cheaper, but hidden costs can creep in."                                 | "Building in-house might seem expensive, but offers long-term savings and IP control."                  |                                                                             |                                                                                                                                     |  |  |                 |
|                                    | Production & Supply Chain                 | "cannot purchase sourcing is done by external agents"                                                                                                                                                                       | seeds are brought in from the companies primed and again send back to them                  | seeds are received from all over the world and then send back after priming                             |                                                                             | seeds are received primed and sent back                                                                                             |  |  |                 |
| Legal & Regulatory Framework       | IP & Legal Protection                     | "patents vs. trade secrets is a choice you need to make"                                                                                                                                                                    | Our priming process is a secret                                                             | they are trade secrets, making patents can risk leaking of the information                              |                                                                             | even our employees are unaware of our priming process                                                                               |  |  |                 |
|                                    | Grant & Subsidy Dependence                | "Grants give startups the breathing room to focus on R&D without immediately worrying about governments and grant agencies de-risk innovation and help with mentoring, networking."                                         | "Grants and subsidies are gold—if you qualify, go for them first."                          | "Some of the most promising technologies I've seen wouldn't have survived without early-stage funding." |                                                                             | what gravity gardens can patent is the specific combination of time and intensity because you cannot patent overall use of gravity. |  |  |                 |
|                                    | Role of Public Institutions               |                                                                                                                                                                                                                             | "Public funding can de-risk innovation."                                                    | "EU programs or innovation hubs have helped many companies."                                            |                                                                             |                                                                                                                                     |  |  |                 |

Figure 13- Thematic Analysis

## **Annex 5 – Interviews**

NUTRILEADS – SUE MCKAY – speaker 3

**Speaker 3 (00:00)**

Hello, I'm Atharva. Thank you so much for taking the time to meet with us today. We really appreciate the opportunity to speak with you and learn from your experiences. We're especially excited to hear your thoughts, since your work aligns so well with what we're trying to build.

**Speaker 1 (00:12)**

Thank you, Atharva. I'm glad we could connect. I'm always happy to support early-stage teams and share whatever insights might be helpful. Looking forward to the conversation.

**Speaker 3 (00:25)**

I'm currently working on my graduation project, and as part of that, we've been exploring an idea where we use gravity as a tool to manipulate plant behaviour.

The concept started from a curiosity about how plants respond to different environmental stimuli, and we thought — what if gravity itself could be a variable we control? We've run some initial experiments and observe different results in terms of growth direction, morphology, and even biochemical responses. It's still early, but the findings have been really encouraging and point toward some exciting possibilities.

**Speaker 3 (00:53)**

So based on that, we get different kinds of results. We found that some plants responded strongly. Now, we are mostly focusing on plants with strong root systems—plants like carrots, radishes—root vegetables.

**Speaker 3 (01:11)**

So, carrots, radishes—those kinds of precision crops. We are also working with garden cress. Mushrooms are not included, but that's part of our future plan.

We spin the plants using gravity-level machines, creating a controlled environment. In garden cress, we saw remarkable changes.

**Speaker 3 (01:38)**

The growth time was cut down significantly. Generally, garden cress takes about 10 days to grow, but in one experiment, we achieved full growth in 3 days. That experiment was very successful.

Of course, many experiments didn't work out, but now we are working with a unique agency. We also have an office at SBI in Norway, near the Space Reservation Center there.

**Speaker 3 (02:10)**

They help us with these projects. Since food can be grown in space, we believe this method has potential. Other companies are also looking into this area. We're collaborating with a professor in Iran.

**Speaker 3 (02:33)**

We currently have about 15–20 engineers: 10 engineers and 5 plant science experts. Robin, our biomechanical engineer, is leading the research.

I handle business-related matters. Paulino is also on the team—he's a mechanical engineer.

**Speaker 3 (03:07)**

Now, we are exploring a new concept: molecular farming. The idea is to manipulate plants to produce certain valuable chemicals—compounds they normally make in very small amounts.

By using hypergravity and environmental controls, we aim to trigger gene expression to boost production of these compounds.

**Speaker 3 (03:42)**

For example, if a plant normally produces 1 gram of a compound, we aim to increase that to 5 grams. So, a 5x increase in yield. That's the research direction.

My part of the research is interviewing companies that are already doing something similar, to understand how they chose their business models.

For example, many companies outsource their production and focus on developing the supplement. So I asked: Why did your company decide to do it this specific way?

**Speaker 1 (04:44)**

We are focused on being an ingredient company. That means we focus on developing the ingredient itself, so that typical consumer product companies (like those making supplements or beverages) can use it in their own products.

**Speaker 1 (05:07)**

We aim to build the scientific evidence and a story around the ingredient—so that others can confidently use it in their own end-products.

**Speaker 1 (05:29)**

We don't do the manufacturing ourselves. Production is outsourced to external partners, but we own the ingredient IP and sell it to other companies who will use it in their products. That's our B2B (business-to-business) model.

**Speaker 3 (05:47)**

Yes, that's what I wanted to understand.

**Speaker 1 (05:48)**

Exactly. It's a business-to-business model.

**Speaker 3 (05:50)**

So the ingredient, like **Benicaros**, is your product?

**Speaker 1 (05:52)**

Yes. **Benicaros** is the brand name. The ingredient is the value.

**Speaker 3 (06:01)**

Does your company rely only on the B2B model, or are there other revenue streams?

As a new company like ours, we're still struggling with cash flow. How do companies like yours manage to keep operations going in the early stages?

**Speaker 2 (06:28)**

You're asking about liquidity for ongoing operations?

**Speaker 1 (06:31)**

Yes, that's always a challenge in the beginning.

**Speaker 2 (06:35)**

Do you rely on fundraising?

**Speaker 1 (06:39)**

Yes. Initially, we raised funds externally. We had investor backing and also self-funded certain research activities.

Clinical studies, for example, are very expensive. So, we work with research partners and apply for grants to cover those costs.

**Speaker 1 (07:07)**

R&D is always a continuous investment. Every 30–60 days we submit applications for funding. We also collaborate with universities and other stakeholders to support operations and share costs.

**Speaker 2 (07:29)**

So that's for funding R&D. What about operations—do you also sell products that are patented?

**Speaker 1 (07:41)**

The company is currently in the **investment phase**. We're focused heavily on R&D.

**Speaker 1 (07:46)**

We want to create strong scientific evidence for our ingredients. Now, we're shifting to **commercialization**—telling people about the research, showing how great the ingredient is, and finding customers who want to buy it.

**Speaker 1 (08:07)**

Even small amounts of ingredient sales can help generate revenue and support fixed costs.

**Speaker 3 (08:15)**

So, just to clarify—

**Speaker 1 (08:16)**

You're asking all these questions because you're exploring funding and business models?

**Speaker 3 (08:27)**

Yes, it's for my **thesis**.

I'm designing a **revenue model** for our startup. We're taking inspiration from different companies—how they survive the early phase, how they scale, and how they structure their business.

We're only a year old and don't have much industry experience, so we're trying to learn from companies like yours.

**Speaker 3 (08:50)**

It's hard here now. Everything is expensive, as you said.

**Speaker 1 (08:57)**

Yes, it is. You need to keep a long-term vision. Decide where you want to be and focus your resources there. Even if progress is slow, just make small, consistent steps forward.

You don't need one big grant to make progress. Sometimes, many smaller initiatives or pilot projects can lead to larger investor interest or funding later on.

**Speaker 3 (09:48)**

Do you think it's better to look for small-scale support or aim directly for large funding?

**Speaker 1 (09:52)**

I can't say exactly what's best for your company. But to move forward, you need solid clinical studies, especially in humans. That can cost more than a million euros – extremely expensive.

There are companies that can help with applications for large grants. It's sometimes better to invest in those companies to write good grant applications for you.

**Speaker 3 (10:55)**

We are extracting carrot pomace at home using water. Where do your carrots come from?

**Speaker 1 (11:12)**

Our farmers grow carrots specifically for juice production. After juicing, the leftover pulp (pomace) is usually waste.

We use that carrot pomace as a starting material. It's sustainable and has health benefits, so companies are interested in our ingredients both for health impact and the sustainability story.

**Speaker 3 (12:09)**

We're doing a small production run in a few days. Can we send you samples? Maybe you can test if it fits your ingredient requirements?

**Speaker 1 (12:40)**

We've already set up a production line in full. Everything runs in a controlled process with fixed steps. If you want us to compare it, we could do it at a lab level, but we no longer have access to smaller R&D setups – our production is scaled now.

**Speaker 3 (13:43)**

We have ties with labs but not for full testing. It's hard to find companies who just analyze products without selling something.

**Speaker 1 (13:57)**

The active ingredient we use is in the carrot's cell wall – it's what gives carrots their structure. If your process enhances that or makes it more bioavailable, it could be beneficial.

**Speaker 3 (14:32)**

Yes, we've seen that stress conditions in plants (e.g., imposed externally) can lead to higher nutrient concentration. We've started some experiments and will test under microscopes with help from our professor.

**Speaker 1 (15:37)**

That's great. But my question is: we are using waste pomace. If growing fresh carrots gives better yield or cost efficiency, I'm not sure if your setup would be more economical.

**Speaker 3 (16:09)**

True, but we aim to make vertical farming affordable – using less land and fewer resources. In India, the main cost is electricity.

**Speaker 3 (16:37)**

Our goal is also to empower farmers. Rather than growing crops traditionally, we want to offer vertical farming systems that could double their production using fewer inputs.

**Speaker 3 (17:12)**

Do you have direct contact with your farmers?

**Speaker 1 (17:22)**

No. We work with the juice factory, and they source from farmers. We can trace back if needed, but we don't contact them directly.

**Speaker 3 (17:41)**

How do IP rights affect your revenue? We're considering applying for a patent, but it's expensive.

**Speaker 1 (18:03)**

IP provides protection and credibility. It attracts investors. If your process is patented, it becomes an asset.

You must decide whether to patent it or keep it a trade secret. Patents are expensive and make your process public, while trade secrets are cheaper but must be tightly protected.



**Speaker 1 (18:58)**

You need to set up internal protocols so only the right people access the sensitive parts. And ensure legal agreements are in place. Otherwise, if someone leaves the company, your secret might be lost or exposed.

**Speaker 3 (19:37)**

That's true. We're seeing similar work from other research institutions, like Japan's space research center. But precision control in our setup is key, and that's not easy to replicate.

**Speaker 1 (20:12)**

Yes, but if you patent everything, someone might improve upon it, and you'd lose your advantage. So, again – strategic choices. Either patent to protect or keep it secret and control access.

**Speaker 3 (21:27)**

We're only 1.5 years into this. How long did it take you to start generating revenue?

**Speaker 1 (21:55)**

It took about 9 years. The first several years were pure research. The last 2–3 years, we moved to commercialization. But again, this depends on your product. Clinical studies, analyzing results, and market launches all take time and money.

**Speaker 3 (22:35)**

Yes, if something goes wrong, the whole product may be scrapped.

**Speaker 1 (22:51)**

Exactly. Our research began at Unilever. A scientist was asked to find a nutrient that had long-lasting benefits. Carrots turned out to be the most promising after a lot of literature review.

**Speaker 1 (23:38)**

We looked for crops that were effective but affordable. Carrots and other root vegetables had the best profile.

**Speaker 3 (23:52)**

We've also done a lot of literature review, but it's hard to draw conclusions. Studies often contradict each other.

**Speaker 1 (24:15)**

Yes, that's why you need to test everything yourself. Validate the hypothesis. Make sure you have your own strong experimental data.

**Speaker 3 (24:52)**

We're focusing on plant-based systems.

**Speaker 1 (24:54)**

Yes, are you targeting specific crops or functions? Like nutrient-rich foods for astronauts or greenhouse crops?

**Speaker 3 (25:19)**

Yes, many companies in India work on that. We're developing machines that can grow crops in challenging environments.

**Speaker 3 (25:52)**

We have two goals:

1. Make local companies more self-sufficient (especially in case of trade disruptions).
2. Make farming possible in extreme climates, like desertified areas.  
We also want to restore ecosystems.

**Speaker 1 (27:04)**

That's a very broad mission. I suggest focusing. If your goal is to restore ecosystems, that's one path. If you want to generate revenue quickly, that's another. You'll need to decide what impact you want in 10–20 years.

**Speaker 1 (28:22)**

Within the company, you'll always have 5–10 ideas. You must choose carefully. At the end of the day, you'll need to ask: what do I want to achieve, and how do I want to get there?

**Speaker 1 (28:33)**

But on January 21st, it will be 15 years since we started. I had a question yesterday about setting up a company. Even if your ambition is idealistic, you still need a rough idea of where you're going and what mistakes to avoid.

Because when you raise money, investors want clarity. They're parting with their money, so they expect a structured plan.

**Speaker 1 (29:00)**

You have to define what the final impact or outcome is. Is it an economic return? A societal change? Investors will ask tough questions – they want to know the financials, the business logic, and the change you promise to bring.

**Speaker 1 (29:25)**

There's a wide variety of motivations. Some want to innovate, some want to make a social impact, others just want returns.

**Speaker 3 (29:32)**

Yes, it all sounds like it matches what we're going through.

**Speaker 1 (29:35)**

You care about the work, and you're a realist – that's essential. Ask yourself: What does your vision look like in 5, 10, or 20 years?

Even if today the idea feels unique, in a few years others might join the space. If too many people work on the same thing, you need to ask: will there still be space for your product in the future?

**Speaker 1 (29:59)**

Is this the right space for you to begin in? Try to focus. Don't get lost in all the possibilities.

**Speaker 3 (30:08)**

Yes, absolutely.

**Speaker 1 (30:10)**

Usually, it helps to have a helicopter view – look at the broader landscape, then build your strategy from there.

**Speaker 3 (30:20)**

Yes, we're doing a lot of work on that.

**Speaker 1 (30:23)**

That's good. Especially at this early stage, it's important to zoom out and decide where you want to go.

**Speaker 3 (30:38)**

It's overwhelming sometimes. We're all over the place because there's so much to do – everything feels important.

But we've realized we need to focus, conclude this phase as a learning experience, and then pick the most useful product. We can use that to generate revenue and fund the next stages, including further research.

**Speaker 3 (31:05)**

We're trying to develop something that can generate revenue and momentum.

**Speaker 1 (31:08)**

That's smart. You don't have to build everything at once. Sometimes it's better to have a front-line product, another in development, and others still being researched.

But you do need a business plan to communicate all this clearly.

**Speaker 3 (31:31)**

Yes, I've been working on a business plan, or more like a revenue model for the first 3 years. Right now, our main income is from grants and investors.

We've been applying for multiple grants, like from SBIR, and we're partnering with educational institutions to do collaborative research.

**Speaker 3 (31:51)**

That's our main source of revenue right now.

**Speaker 1 (32:00)**

That's a good route. Partnering with research institutions is a strong strategy – they're often really interested.

Also, there are experts who can help you with grant writing. You'll have to pay them, but they can greatly increase your chances of success.

**Speaker 1 (32:27)**

Sometimes they'll even help you frame the proposal or bring in new ideas.

You don't always have to do everything yourself. Just make sure you stay involved enough so your team gets hands-on experience and builds knowledge.

**Speaker 3 (32:47)**

Do you export your products?

**Speaker 1 (32:50)**

Yes, we sell them. We're already in the market.

**Speaker 3 (32:59)**

To Japan?

**Speaker 1 (33:01)**

Yes, Japan. And we recently launched a product in Denmark, though it's currently paused due to some issues with distribution.

But yes, we have multiple products in different markets.

**Incotech**

Interview with Rob Pronk – Global Marketing Manager and Global Regulatory Manager, Incotec

Q1. What are the most commonly used seed treatment technologies? Since Incotec is known for seed priming, is seed priming more popular than seed coating, or what's the case?

A: That's a very difficult question. You need to understand that at Incotec, we have two very different business models: vegetable seeds and field crop seeds. With vegetable seeds, our customers send their seeds to Incotec. We prime and pellet them—for

example, lettuce or tomato—and then send them back.

With film coating, we produce the coating and send it to the companies, and they coat the seeds themselves. For crops like maize, soybean, and sunflower, it has never been done at our facility—customers handle that themselves.

So, whether we're talking about field crops or vegetables, the scale makes a difference. If you measure in terms of acres, much more seed is coated than primed, because field crops cover hundreds of millions of acres. Vegetables are planted on a much smaller scale, and priming is a niche sector. So, yes, film coating is used much more than priming.

Q: Is seed coating so easy that your customers can do it themselves? Can farmers coat seeds too?

A: Our customers are not farmers; they are large seed companies, and they have their own facilities to handle coating. Farmers usually don't coat seeds themselves—it's very rare. The only place where that sometimes happens is Brazil.

Q2. In your opinion, to what extent does priming affect seed yield, quality, or profits for farmers?

A: I would say, in general, that priming doesn't increase yield directly, but it improves the efficiency of the growth process significantly. What it does is broaden the germination window of the seed.

In Europe, all vegetable seeds are transplanted—you take the seed to a nursery, grow seedlings, and then plant them in the field. Onion seeds are an exception. They aren't usually primed because priming them is too expensive due to the large volumes required.

Q3. What are the main challenges your customers face—logistics-wise, supply-chain-wise, or cost-wise?

A: It really depends. Some crops benefit significantly from priming, while others don't benefit much at all. It also depends on how easily a seed germinates under normal conditions.

For example, onion seeds benefit a lot from priming, but the costs are too high relative to their market value. Apart from that, I don't see major issues for our customers.

Q4. Can you tell me about pricing—how much does it cost to prime a ton of seeds?

Or, if I have a batch of unprimed seeds, how much would it cost to prime them?

A: When it comes to vegetable seeds, pricing isn't based on weight—like tons or kilograms. We price by the thousand seeds. Depending on the type of seed, our priming charges are between 20–30 cents and up to 1 euro per 1,000 seeds.

But with that, we guarantee that our customers gain more than they pay because they can sell primed seeds at a higher price than regular ones.

Q: Is that charge enough to run operations and still make good profits? Honestly, 20 cents seems a bit low.

A: Yes, it is profitable—we make a lot of money from it, no doubt. Otherwise, we wouldn't do it.

Q5. What do you think of new trends and innovations in the market? What will the industry look like in 5–10 or even 20 years? Do you think priming can be replaced?

A: Good question. I believe the market will still be there. Breeders will always look for ways to improve seed quality and germination. But they have many parameters to manage, and they can't address them all at once.

The need for primed vegetable seeds will continue to grow, and we'll become more efficient in how we provide them.

Q6. What are the most common seed priming techniques—for Incotec, your competitors, or the market in general?

A: To be honest with you, I don't know—and even if I did, I wouldn't be able to tell you. Our priming techniques are kept secret, even from internal employees.

Let me tell you a funny story. In our priming process, we use certain ingredients labeled under code names like C-22, B-18, etc. The people who handle the chemicals don't know which process they're used for, and those who assign the codes also don't know the exact usage. Our priming facilities are separate and restricted. So, when we lose staff, nobody can fully replicate the process. This might be the case for many other companies as well.

We can't patent it, so we keep it secret.

Q7. Let's say I'm an entrepreneur who wants to start a seed priming company in the Netherlands or Europe. Would I be able to succeed? Can I be profitable?

A: No, no, no—it would be extremely difficult to enter the market.

The seeds we prime are very high-value, and seed companies are very cautious about quality. It's very easy to mess up a batch, and there are thousands of ways to prime incorrectly. Learning what works takes time, experimentation, and analysis.

Also, the market is very locked and niche. Gaining the trust of seed companies is extremely difficult. If a batch fails, the seed company can be held responsible for massive crop losses, which makes them very careful about who they work with.

Q8. From what I understand, your customers are big seed companies. You prime their seeds and send them back. Has Incotec ever considered producing its own seeds, priming them, and selling directly to farmers? It seems like it could be a good revenue stream.

A: No, no—we don't do that. We're not a seed company. Seed companies breed, grow, and manage all aspects of seed production.

If we started producing seeds, we'd become direct competitors to our own customers.

That would ruin our business. So, no—we have no interest in doing that.

### **Conversation with Jeroen Kuin sales manager at de groot n slot seeds**

Atharva: Thank you for taking the time to speak with me today, Jeroen.

Jeroen: Of course. It is my pleasure.

Atharva: I have been quite intrigued by the seed priming process and its application across various crops. I understand that your organization has conducted considerable research on this, particularly for crops like wheat and field crops. I am curious to know whether similar results have been observed for other crops.

Jeroen: I appreciate your interest. However, my responsibilities lie primarily within commercial operations, and while I have discussed this topic extensively, the specific details regarding scientific research and field applications fall under the expertise of our technical specialists. I would recommend consulting with one of them for a more accurate and informed perspective.

Atharva: That is completely understandable. Thank you. To proceed, I'd like to inquire more about your company's focus. We primarily deal with allium crops such as onions and garlic. Has your company considered expanding into vegetable or field crop seeds?

Jeroen: Our company remains firmly committed to allium crops. We have maintained a successful partnership with Bejo Seeds for over 55 years, and our focus on allium remains our core strength and

market differentiation. Our partner, Bejo, is actively involved in a wide range of vegetable crops, but we dedicate all of our resources to garlic, shallots, sweet onions, onion sets, and other allium-related products exclusively.

Atharva: I see. Does your company conduct seed priming at your own facilities, or is this service outsourced?

Jeroen: All of our seed priming and seed treatment activities are conducted by Bejo Seeds at their facilities. While we produce many of the seed parent lines and develop the varieties, all subsequent handling—such as priming, seed cleaning, and treatments—is performed at Bejo's location. Although it is not physically located within our company, the work is done in close collaboration, effectively making it an in-house operation for us.

Atharva: Understood. I have read that plant breeding is a major component of your company's operations. Since seed priming enhances germination speed and quality, is there any potential to achieve similar outcomes solely through breeding, thereby eliminating the need for priming?

Jeroen: This is a thought-provoking question. In fact, our management team discussed this during strategic planning sessions last year. Theoretically, the objective of breeding varieties with inherently faster and more uniform germination is desirable. However, based on our current knowledge and regulations, we do not foresee replacing priming through classical breeding methods. Genetically modified organisms (GMOs) might offer possibilities in that direction, but GMO use is not part of our breeding strategy in Europe.

Priming plays a vital role not just in germination speed, but also in ensuring storability. Seeds that are partially germinated—through priming—require highly controlled storage conditions. Relying entirely on breeding for such traits could compromise seed viability over time, which is crucial for our role as a reliable supplier.

Atharva: Thank you for the clarification. I have found it quite challenging to identify the size and scope of the priming market. Do you have any insights into the extent of seed priming adoption?

Jeroen: The adoption of priming varies significantly by region. In the Netherlands, for instance, approximately 65% to 70% of seeds we market are primed. This figure can reach 100% for certain varieties. We are also observing rapid growth in countries such as Germany, France, and across Scandinavia—where short growing seasons make rapid germination essential.

Atharva: That is indeed a remarkable penetration rate. Do you foresee any developments in the near future that might serve as alternatives to priming—perhaps methods that could reduce costs for farmers?

Jeroen: We are continually exploring more efficient and cost-effective solutions. At present, however, there is no viable technique that can entirely replace priming. Should a promising alternative emerge, we would certainly evaluate and potentially adopt it, provided it aligns with our standards for quality and reliability.

Atharva: Could you share the cost structure for priming, particularly as it applies in Europe?

Jeroen: While I do not have precise production cost figures, I can share that we currently charge €60 per unit, where one unit comprises 250,000 seeds. This price is subject to annual review and is influenced by fluctuations in labor and operational costs.

Atharva: Thank you for the insight. I have encountered literature discussing the benefits of priming in field crops such as wheat. Despite promising research results, priming is not widely used commercially in such crops. Is cost the primary deterrent?

Jeroen: Yes, cost is the principal barrier. In crops like wheat or maize—large-scale, low-cost field crops—the economics of priming are far less favorable compared to high-value vegetable seeds. As a company specializing in allium, we do not focus on field crops. For such topics, organizations like KWS in Germany—who are heavily involved in field crop breeding—would be more appropriate contacts.

Atharva: That makes sense. Do you currently export seeds, and if so, are those seeds primed and processed in the Netherlands prior to export?

Jeroen: Yes, we export extensively—to China, the United States, Africa, and other regions. At present, all seed handling, including cleaning, treatment, and priming, is centralized in the Netherlands. However, we are actively working to establish additional regional facilities to support these activities locally, which will enhance both efficiency and sustainability.

Atharva: That seems logistically complex. Does it not result in significantly higher transportation costs?

Jeroen: Indeed, there are logistical challenges and associated costs. Nevertheless, quality assurance is paramount. By processing seeds centrally, we maintain full control over quality and consistency—an essential part of our value proposition. As our reputation is based on the quality of our seeds, this is a non-negotiable element of our operation.

Atharva: Your commitment to quality is commendable. Given your long-standing success, I'm curious—where do you believe the company still faces challenges, and where is there room for improvement?

Jeroen: That is a thoughtful question. While we have a strong market position in long-day varieties—particularly in Europe and the UK—the short-day segment remains a growth area for us. This includes markets such as India, where hybrid adoption is still developing. We are now investing significantly in these regions, aiming to expand our portfolio and presence. The evolution of local agricultural economies is encouraging, and we expect hybrid adoption to increase steadily in the next 10 to 15 years.

Atharva: That aligns well with recent market trends. Thank you so much, Jeroen, for sharing these valuable insights. I truly appreciate your time.

Jeroen: You are very welcome. It was a pleasure speaking with you. I wish you the best in your studies and professional endeavors.

Atharva: Thank you. I am currently pursuing my degree in International Agribusiness and Trade at Van Hall Larenstein. I previously completed two years of International Business Management in India, and I am now completing my final project and thesis here.

Jeroen: That sounds excellent. I wish you every success. Should you have further questions, please feel free to reach out.

Atharva: Thank you, Jeroen. May I also connect with you on LinkedIn?

Jeroen: Certainly. Please feel free to send an invitation.

Atharva: Thank you once again. It has been a pleasure.

Jeroen: Likewise. All the best.

**interview with Eric Jukers CEO Centor Europe (Netherlands)**



## Page 1

### RELATIONSHIP

↓

Last independent seed company in the world

Competitor (Amotech, Germans etc.) → Competitors on certain segments

### INDEPENDENT & ADVANTAGE

**DIFFERENCE** → Small facility → More about helping customers

Shorter logistics (process)

They help customers choose the best technology

↓

**Ideal for gravity coating!!**

Their client base is chemical or seed companies

↓

Farmers don't know what they want or need

↓

Farmers do not know exactly

### (CENTOR)

- They are a B2B
- Don't work directly with farmers
- They do that in 3rd level countries

## Page 2

It's already been done → Seed companies try to create awareness for priming

**Not exactly "known"** → The price of priming

↓

But very **crop dependent**

It's a question of scale → Field crop seeds are just too expensive to dry after priming

↓

Scale of equipment is too big

In certain cases you

**cannot substitute priming**

↓

Because you bring maximum level at the same time

↓

Because priming activates metabolism (due moisture)

And coating does can't do that

## Page 3

If seed companies are smart

↓

They will not allow genetics to take over the purpose of coating & priming

↓

DNA is like a hard disk

If you add something, you remove something else

↓

Only exception is GMOs

↓

Divide the tasks between sectors

## FULL TRANSCRIPT OF CENTOR CONVERSATION

**Eric J:** [00:00:00][...] We being independent? Of course, that has a that has a value to certain companies in this world, especially the larger companies. And I mean, we are international as well. So we cover the globe. So we compete with a lot of these bigger names, which I just mentioned as well. We are different as well in that we have a philosophy. If you come to our offices on the on deck, we really have a small facility. We don't have a big factory because we don't want to get seen in. We help implement technologies at our customer sites all where the seed is produced. Why do we do that? Because we believe that's where you need to to be. And that creates or that stimulates a shorter logistic process because day to day, I mean seniors produced if you take, let's say 20 years ago, seed would be produced, it would be clean, it would be processed, it would be put on the shelf and it would be sold only the season afterwards. Because they couldn't get that all in time. Day to day people don't have that time anymore. And you don't want to have seen on the shelves because that is that's actually lost money. That is dead money. So they want to harvest the seed, they want to process the seed, and they want to sell the seed before the next season. That gives them anything from 3 to 6 months window in which they need to do all these things. If you need to ship seed from China to Europe to start the processing process, you're already six weeks to eight weeks. That means two months logistic travel time. Now you need to get it inside. You need to hope that all the quality control tests work out well, otherwise the seed may be blocked.

**Eric J:** [00:01:53] Then you start the processing process. You need to do the packaging, and you need to get it out in time because harvesting usually starts in in June, July, August or from August September. Seed usually comes into the factories they are processed, but from October they need to go out already to the farmers and they need to be made ready for selling because, I mean, seed companies don't always sell directly to farmers. It goes from a seed company to either a distributor or a cooperative. And, you know, so there's still a whole logistic process behind it before it reaches the farmer. And then the farmer requires it by January, February, because as soon as the time is right, they want to start seeding. Or in the case of vegetables, they need to be at the plant in December

because these tomato plants need to be created and they take 6 to 8 weeks. That means that they need to they can only be sold to the farmers around February. Right? So, you know, that whole time span that's very, very short eventually. So what we do is we, implement the technology where it's needed. So if seed is produced in France somewhere, we would implement the technology at the seed production facility so that they produce the seed. They do a pre-cleaning with a cleaning step. They can package, they can treat, and they sell it directly off to the seed company as a final product. So the seed company does not need to bring seed here and then send it over to another company to be treated and know everything is done at one spot.

**Atharva :** [00:03:27] That's very interesting because it's sort of like a solution to this globalization phenomenon, which of course has touched also the seed industry. But yeah, of course, it's much better to instead of having it treated in different places where it might be. How can I say consolidated that it's this way or it's cheaper to do it this way, but it's you have to sacrifice time for it. But then, of course, if you can do everything in the same spot and you can just snap your fingers and you have your product. Yeah, that that helps a lot. That's a very interesting because that's more specific in terms of if this is what you do as a company, this is a very specific service. But I feel like even though it's very specific, it's a very needed and appreciated and, and it it differentiates you from the larger companies because as you said, those are more chemical companies sort of with a specialization in seeds. And whereas you guys are really focused on seeds and you do something that nobody else does, which is facilitate logistics. And how do you say save time?

**Eric J:** [00:04:37] So we facilitate we don't facilitate logistics. We facilitate making the whole logistical process more efficient. Right. The company actually does that in the end, but we help them I mean we we come up with that idea and we help them implement that idea and make it for them more efficient. And it always works. We see that. So we work very closely together with machine building companies as well that we then again introduce to companies and they can, they can, let's say build the factory or build the machines within the factory and help do that process. And the only thing we do is we just focus on our seed coating. And what we do do is we do give training, we do do audits, we do all that kind of thing. So and we also help customers because we are very objective in that respect. We help customers select the best technology. So when they come to us and they say, okay, you know, we have so many sunflowers coming through the process. And we need to increase those sunflowers, for instance what type of machine would be good for that? So we have a look at okay, I have a crop. Sunflowers. Do they have any other crop they want to treat as well? Yes or no.

**Eric J:** [00:05:50] We take that into account. And we say okay for sunflowers and encrusting, you would need a machine that has this capacity because you need to do it within 3 or 4 months. We look at the technical details. You need powder feeders. You need so many pumps. You need this, you need that. And then we say, all right, you know, we have 2 to 3 machine companies that we can present that we know have good quality products or good quality equipment that is able to do this and able to handle it. And very often also with whom we have results already. So that means that they can take the machine, they can put the machine in the factory. And as soon as the machine is started up, we already have a recipe that we know works, and we can immediately implement that recipe for the customer. We give the operator training and we also help with the quality quality testing if needed. Right. So all that type of thing. So and then we implement it and we only leave as soon as things are working and running well.

**Atharva :** [00:06:56] Once you have created a system.

**Eric J:** [00:06:59] Correct? Correct. And companies really appreciate that because in the past, if they were to invest in a machine technology, they would actually go to a machine company. And every machine company has its usp's. And based on the usp's you are presented, you need to make a decision. And very often companies only look at the bottom line as well. And they say, you know, there's a difference between \$10,000 between the one and the other. We'll take the lower one. Yeah. Because I mean, why do we need to pay the extra \$10,000? But having us involved in that, we can we can tell you or give you a more objective perspective on what the differences are, because we don't look at the usp's from the companies. We we want the customer to obtain the result, the desired result. So the customer has a requirement, he is looking for a specific result. And we can give him more that the vision from if you take this machine, you will be able to achieve that result. And we can show that to him, because very often, as I said, we already have a history. We know how the machines work and we've already done these products on the machines.

**Atharva :** [00:08:06] So yeah, that must be a very valuable service. I mean, it's at the end of the day, you pay for a level of expertise and the kind of consultancy that you guys have with your your experience. So I would assume that for a farmer or for a company, it means a lot because they they don't have to make the choice themselves without how do you say the proper education or experience in it? But they can trust you guys.

**Eric J:** [00:08:31] Yes, yes. And very often, I mean, we it's interesting because you use the word consultancy. I know there are companies like in tech, and so you could get them in and they would come in as a, on a project basis. They would say, okay, this is for us. This is a project and we will work on it as a project. And they will they would charge a grand fee for that. We don't do that very often when we are starting a discussion with the company. Our first criteria is to evaluate how serious this customer is. If we realize that the customer is not serious, we will we will speak. Let's say our truth and we will tell them. We don't think that they're serious and we are not willing to cooperate. So we only cooperate with serious, serious partners, people that are serious in implementing technology or setting up a new factory to their benefit. And then we start our discussions and based on on the size of the customer, based on the level of interest, we will then decide, okay, are we going to charge them for a fee? Very often if we need to make large costs, like for instance, if we need to travel somewhere or that type of thing, we will charge them for the travels and the time. But we basically, we don't charge exorbitant fees or we don't charge fees for the service, because for us, it's a service. Because we know in the end, when we implement the technology, even though it takes two years, they will take our products.

**Atharva :** [00:09:58] Yeah, yeah, yeah.

**Eric J:** [00:09:59] And for us, the goal is eventually to get in there with our products. I don't want I don't want them to pay consultancy because that's one time, for instance, for a whole project like that, I could maybe earn \$50,000. But I'd rather make \$100,000 annually.

**Atharva :** [00:10:15] Yeah.

**Eric J:** [00:10:15] On the products that I sell to them.

**Atharva :** [00:10:18] Yeah.

**Eric J:** [00:10:19] Instead of that one time 50,000. So we have a different philosophy in that.

**Atharva :** [00:10:24] That's That's very nice. That's definitely very nice. It feels like a honest business, in a way. In a good way for sure.

**Eric J:** [00:10:35] Well, yes, but that's that's something that's high on our list as a company. And also the people working within this company. That's honesty and integrity.

**Atharva :** [00:10:46] Yeah.

**Eric J:** [00:10:47] That's that's that's number one for me, because that is how people know me within the industry, because honesty and integrity mean that people trust you. You say. Let's say you mean what you say and you also do what you say. I can. I can always back that up.

**Atharva :** [00:11:06] Yeah yeah yeah, yeah for sure, for sure.

**Eric J:** [00:11:10] And as I say, sometimes I need to tell the customers. Know what you want cannot be realized. Or if you really want to go through with it. I'm not going to work with it. I'm not going to. I'm not going to assist you.

**Atharva :** [00:11:21] Yeah. Yeah. I mean, that's got to be part of the job at the end of the day. May I ask would you say in your experience do farmers or seed companies is there a higher demand for technologies on seed coating or on technologies for seed priming?

**Speaker3:** [00:11:43]

**Eric J:** [00:11:44] Okay. Just just to clarify one thing. We we are a B2B company. So And within the industry, we have an agreement that we as technology companies do not work directly with farmers.

**Diego:** [00:11:55] Okay.

**Eric J:** [00:11:57] Basically, that's true for Europe and America. I think as soon as you get to, let's say, the second or third world countries, like in Africa or in South America, where you have large growers, we would have the exception because then you would be on a grower level. But it's because there is no other option within the countries that they are present. But basically we only deal with our customer base is seed companies, seed cooperatives, seed production companies or chemical companies. We don't directly work with farmers.

**Atharva :** [00:12:31] Okay.

**Eric J:** [00:12:33] If you look at the farmer level in the farmers requirements, what do they require? Very often the farmer does not know. That's a it's a question again, of very often a farmer likes a specific variety for some other reason, or he's seen it performing, or he's got the experience that it performs. So variety is number one. And secondly, very often he is guided by either the chemical company or the seed company for the specific treatment he requires. And why I say that is because you see that these days, specifically in Europe, where we're not allowed to use certain chemicals anymore. Everybody sort of up in arms and don't they don't know what to do. Even the farmers don't know what to do. And again, they need to rely on seed companies or other companies that actually try and sell them biostimulants biologicals or soil treatment materials to try and get the crop healthy and performing. And I think your question was also, you know, the difference between seed coating and seed priming. So what is the requirement? Again? Game. Really? My honest opinion. I think farmers do not know. And we come back to the start of our talk. That is the unknown of technology.

**Eric J:** [00:14:01] A farmer may may have heard of a priming, and he may have seen the effect of the priming specifically when you speak about vegetable open field crops. But it doesn't mean that he understands it. So they basically go with what their experience is. If he's, he's if he's experience is good, then in his region when he uses primed onion seed that it works fantastic. Then it's specific for that farmer for that region. And he's experienced because he may be living in a in a region where, you know, when the when the rain comes then he needs to sow before the rain comes. The rain comes and he gets a very quick germination because he may have a type of soil which forms a thick crust after heavy rains have passed through there. And they are very important parts in northern Italy where that actually happens, where especially now when you have these thunderstorms come through, they actually compact the soils on top of the sea. If you do not get a rapid germination while the soil is still wet, that means that the seed cannot get through because then the thrust becomes too. Too thick.

**Atharva :** [00:15:14] Yeah.

**Eric J:** [00:15:16] So again, it's a little bit the farmer's own experience. The area where he is is farming the crop that he's working with, etc.. Tomato seed. Tomato seed always goes to a plant raiser is we still have a couple of parts in Europe, I think in Italy and in Spain. In Greece, there may be still areas where they see directly, but I doubt it. I think every thing goes through plant raisers. When you put it through a plant raiser you should actually prime the seat, because what you get is the seed germinates at the same time very rapidly and very uniformly. So eventually when you bring that trait to the farmer, he has a uniform set of plants. You don't need to sort the trade.

**Atharva :** [00:15:58] Right.

**Eric J:** [00:15:59] If you do not do a priming, you know, the first seed germinates after three days, the last seed germinates after 28 days. For instance, you get a big variance in sizes. And then very often you need to clean the trays what they call clean the trays. So you need to repack so the smaller plants go into smaller trays. So you try and resize it. That's a lot of hand labor. Yeah. In Europe we can't afford it.

**Atharva :** [00:16:22] Yeah exactly. Yeah. It makes sense.

**Eric J:** [00:16:24] So when you prime it, you get a uniform product and you get uniform. So let's say plants, you can get to the farmer and we know with priming as well is you can keep that effect so that uniformity until the end. That means when they harvest, for instance, a cauliflower farmer or a cabbage farmer. When they harvest, they can harvest in one go.

**Diego:** [00:16:46] Right.

**Eric J:** [00:16:47] If the plants were irregular, that means that they will need to go in. They would need to sort at the end of the week. They would need to go through again, take the second lot, and then go for a third time. Go through them. That's very, very inefficient. So by doing a priming you can get a much more uniform result in the end.

**Atharva :** [00:17:05] So priming has a beneficial effect as you just pointed out. But again it is not a really well known by many farmers. So do you think that there's an opportunity for farming companies or anybody involved in the agri food sector to start a campaign, to start informing the farmers so that these techniques are going to start to be adopted much more frequently, not only to increase the farmers revenue, but also to increase the production in Europe.

**Eric J:** [00:17:44] I think that's already been done and that's already being done by the seed companies. So the seed companies try and promote that. And why do they do that is because they want their products to perform better than the competitor product. And if we have a look at for instance, cabbages or tomatoes or whatever, all the varieties out there are very, very similar. There are slight differences from, let's say, certain resistances which have been built in. But but if you look at tomatoes, tomatoes originate, for instance, out of seven originally out of seven lines. So all the tomatoes in the world come back to the same seven lines. So genetically, there's very little difference between all the tomatoes out there. So if you take a Roma tomato, for instance, and a variety from or a variety from Red Swan or a variety from Enza. They're very similar. So what are you going to distinguish? How are you going to distinguish those varieties. So seed companies come and they the one does a priming. And that means that it germinates faster. It's more uniform stand and that you have a more uniform harvest. So that's how they would distinguish them from the competitor. But then of course the competitor will always also after a year or two come and they will also start priming.

**Atharva :** [00:19:10] Yeah, exactly.

**Eric J:** [00:19:11] So yeah, the let's say the the drawback for the farmer is, is the seed just becomes more expensive.

**Atharva :** [00:19:20] Oh yeah. Because you have to pay for the priming.

**Eric J:** [00:19:23] Yeah. That's a value added. Yeah. It's a benefit that they give. And the benefit is translated to farming level. So no the seed companies know how to calculate.

**Atharva :** [00:19:33] Yeah I can imagine. Yeah. So would you think like a very ballpark value. How much would you think? The farmers would have to pay more for a prime seed?

**Eric J:** [00:19:49] Oh, to be honest, I do not know, because I'm not privy to. And we're not linked to seed to a seed company. I have no idea what what? They ask for a prime. And I think it's very crop dependent. You know, tomato, especially if you look at greenhouse tomato. It's a high value crop. You pay about anything from \$1 to \$2 or two and a half dollars per seed. So you can do much more to, let's say, increase that value of that seed if you take a cabbage seed that's very cheap. So there's much less you can actually do with a cabbage seed to increase the value of that. So it's very dependent. I don't think there's a, there's a fixed margin that you can say, you know, they put a 50% margin or a 100% margin on that. It depends. Depends on the type of problem.

**Atharva :** [00:20:44] Yeah I wouldn't yeah.

**Eric J:** [00:20:45] You want to. What you can say is look, agronomical crops are basically not primed. The volume is just too big because then you're speaking about hundreds of kilograms of of crop that needs to be done. And the last seeded vegetable varieties are not primed. It's not worthwhile like beans, peas, etc.. The seed has a different value. Plus the the bulk volume of that is so enormous that it's very difficult to do that. So very often the small seeded vegetable varieties are prime tomato cabbages eggplants onions, carrots. They are they are prime. And you see the benefit of that. And of course, the priming is not just only to get, let's say, the uniformity. Sometimes the priming is used to break dormancy. Yeah. Take lettuce for instance. Lettuce is thermal dormant. So they prime it so that it germinates at a broader temperature range. So you can plant it earlier in the season Or you can plot it in regions where you could previously not plot it. Like for instance what is it? I wanted to say not drought conditions. What is the Sahara called in English?

**Atharva :** [00:21:55] The desert, the Sahara.

**Eric J:** [00:21:56] The desert. Thank you. I was lost for that word. They plant. They plant lettuce in desert areas. Especially. You see that in in the US, in the Utah, Arizona. They have large lettuce fields there because they've been able to broaden, let's say, the temperature range by which that lettuce can actually grow.



**Atharva** : [00:22:16] Interesting. I didn't know that. Yeah. And is there so is the barrier to adopting priming more as a technique? Is it purely a matter of awareness and and yeah. Publicity or is it also is there a financial limit? Maybe because we were mentioning that it's feasible for vegetables and it's unfeasible for field crops and field crop seeds for a matter of space and volume and and convenience. But I think I read online somewhere also that sort of the when, when it comes down to field crops, the added value brought on by priming is not worth it. And this particular paper was saying that it is not worth it because in most types of priming, you would have to put the seeds in a solution and it's expensive to dry them out. So unless the crop is very valuable, like for vegetables, if you do that for, you know, cereals you would have to spend so much money drying them, drying all of these many huge quantities of seeds that it becomes unvaluable and valuable as a practice.

**Eric J**: [00:23:36] Correct? Yes. It's a it's a question of scale. If you look at tomato, for instance, tomatoes are fairly small seed. There's around about 300,000 seeds per kilogram. If you compare that to maize in maize, we have about one kilogram of seed is about 4000 seeds.

**Atharva** : [00:23:57]

**Eric J**: [00:23:58] So it's 100. What am I saying? 100. 100 times difference or something of that sort. So that means that when you look at equipment, when you want to prime the same volume of seeds, the scale of equipment need become so big that it's not worthwhile anymore. It becomes too expensive.

**Atharva** : [00:24:17] Right. So

**Eric J**: [00:24:23] That's that's that's, you know, that's where the difficulty comes in. But what, what companies are looking at and I mean, that's something we do as well. So we look at it differently. So we can prime a vegetable seed to get, let's say the benefit out of that we would look at what can we put in the seed coating to benefit agronomical crop to try and get that same effect, or at least better effect so that the seed grows more rapidly, becomes more stronger that type of thing.

**Diego**: [00:24:56] Right.

**Eric J**: [00:24:56] So, so that's, that's what we would look up in Agronomics.

**Atharva** : [00:24:59] Right. So basically you think, okay, what's a solution to have still similar or same beneficial effects while not paying as much money? And a lot of times I'm assuming this answer is coating instead of priming for many crops.

**Eric J**: [00:25:14] Correct? Correct. Yes.

**Atharva :** [00:25:16] And do you think in your experience that maybe in the next 20 to 30 years there will be some innovation in the technology that will either allow priming to become how do you say affordable? Or maybe there will be a new treatment, a new seed treatment technology that will be much more efficient, something completely new that will substitute substitute? Yeah. Priming or the coating all together.

**Eric J:** [00:25:48] You will in a certain case, you will never be able to substitute priming, depending on what the requirement is for the priming. For instance, if you do a seed production, take, for instance, onions or, or carrots. The way that the flower is, is carried the seeds that you have harvest from one, let's say combined flower, because they have a combined flower. A combined flower never starts to flower at the same time. For mine, flower, of course, consists of many flowers in, in it like a sunflower as well. They never ripen at the same time. That means that the seed formation starts at certain flowers and then expands to the other flowers until you have, let's say, the that whole crowd, which is filled with seed that is eventually harvested at a certain time, and then the seeds are taken out. So within that seed population, from that one flower, you have mature seeds and you have immature seeds. What you do with the priming is you expose this, let's say this seed population you expose to the priming. And within the priming you bring the maturity level to the same level. So the mature seeds stay at that level and the immature seeds become mature to the same level.

**Eric J:** [00:27:07] That's what you can do with the priming. That's why as a result you get a uniform and a faster germination because now all the seeds are on the same maturity level. So when you put them in the soil, they germinate at the same time. And you know, they grow and they all very uniform because they're all at the same maturity level. That's of course, one thing a priming does. A priming, as I said to you can also let's say overcome dormancy. We had photo dormancy or germination, dormancy or whatever dormancy there may be. You can also overcome with a priming. Now with a seed coating, you will realize you can never bring the seed to the same maturity level. Because basically, in order to do that, you need to activate the metabolism of the seed. You can only activate the metabolism of the seed when you increase the moisture content, because seed becomes active when they have enough moisture and when there's oxygen available. And that's something you cannot give with a seed coating. Do you follow what I'm saying?

**Atharva :** [00:28:17] Yeah.

**Eric J:** [00:28:17] But what you can do with the seed coating is you can add certain elements or certain biologicals or certain bio stimulants to the coating in which you do as soon as the seed starts to germinate, or as soon as the seed is in an environment that it takes up water, and these elements can go in inside the seed and they can overcome the dormancy, or they can stimulate the plant to overcome dormancy. Yeah. So that's something you can do with the seed coating.

**Atharva :** [00:28:43] Yeah. I was having this example when my supervisor at the university the other day and I told him this. It's a, it's a very like it was a way for it's a classic way for students to oversimplify things, but it helps me remember, I told them I'm starting to feel like priming is like giving a human food, because you stimulate their metabolism and you make them start to grow. And coating is like giving a human steroids, but you cannot give a human steroids without feeding them, you know, because the steroids, they give them things to make them perform better, you know, like run faster or

be stronger, but it doesn't start to make you grow, let's say, in the sense of going from age of five to age of 20 growing in that sense.

**Eric J:** [00:29:36] Yes, correct. So you can by by a priming by feeding the babies, they become bigger.

**Atharva :** [00:29:44] Yeah.

**Eric J:** [00:29:45] But adding steroids, if you give steroids to a baby, he's not going to become bigger. Exactly. Well, in a sense, he is. But he's going to become a michelin man. Yeah. And if you feed steroids to a man, he's also becoming a michelin man. But they're not going to be they're not going to perform similar because the baby's still going to be a baby. And the, the, the man is still going to be a man.

#### **notes from Germain's interview**

PATRICK ↓ PLANT SCIENCE NINJA PhD ↓ WORKED @ MONSANTO, MARKETING 30BS LEAD SCIENTIST NOW ↓

WHAT ARE THE THINGS THAT INFLUENCE SHAREHOLDER VALUES AND THE PREMIUM/QUALITY PRICES? 3 REVENUES NEED TO BE MET

PRICE OF TREATMENT IS RELATED TO PRICE OF SEED ITSELF + WHAT'S VALUE OF SEED TREATMENT?

5% margin NOT ENOUGH ==> 30% TO 60% YOU CANNOT LOOK AT IT IN A GENERIC WAY DEPENDS ON SPECIES & TYPE OF GROWTH

#### **other**

DEMANDS FOR SERVICES ARE ALWAYS DIFFERENT SOMETIMES THEY AIM TO INCREASE YIELD, SOMETIMES DROUGHT OR PESTS...

NOT BIG DISTINCTION IN TREATMENTS OF SEED INSIDE OF EU SALINATION OF SOILS => SEED TREATMENT INNOVATION

GERMANY -> DOES ONLY PRESENT TREATMENT PRODUCTION -> THEY PARTICIPATE IN THE R&D ACTIVELY

AT THE MOMENT NO CURRENT EXPECTED GAME CHANGER, BUT IT HAPPENED IN THE PAST BEFORE

PRIMING & COATING WILL BOTH PLAY A BIG ROLE -> FUNDAMENTAL FOR CERTAIN PLANTS (SMALL)

CHEMICAL COMPANIES DON'T TAKE THE RISK OF DEVELOPING NEW CHEMICALS CUZ THEY GET BANNED SOON

HARD TO PREDICT WHAT REGULATIONS WILL DO 2 WAYS <- NEED TO ALLOW TECHNOLOGIES COMPETITION LEADS TO LIBERALIZATIONS

Transcript – Ravi Kumar and expert from NASA(former), now a professor at UAS, Dharwad.

**Speaker 1:** How do you evaluate the performance of seedlings after priming? Are there any standard protocols?

**Speaker 2:** Good question. I mainly work with wheat—it's my model crop. The seeds are large, and the shoots grow quickly. In 8 days, you can get seedlings up to 10–12 cm tall. I usually measure shoot and root length, leaf greenness, number of branches, and biomass—both fresh and dry weight.

**Speaker 1:** Do you use any advanced instruments or is it more manual?

**Speaker 2:** Mostly manual. Just a regular measuring tape for length. It's simple and it works. Germination is done on filter paper, then after 8 days, we transplant to pots filled with local soil and move them to the greenhouse. That's where we observe further growth—it mimics field conditions.

**Speaker 1:** That's a practical and scalable setup. May I ask—how did you end up working on this topic?

**Speaker 2:** Sure. My PhD was actually on fruit flies, focused on neural plasticity. I was selected for a NASA Postdoctoral Program fellowship. I submitted a proposal, it was approved, and I ended up spending 7 years at NASA working on plant stress under microgravity and hypergravity. Later, I chose to return to India to work on agriculture—closer to home and more directly impactful.

**Speaker 1:** That's really inspiring. Is there any possibility of collaborating with you or your lab?

**Speaker 2:** Definitely. I'm open to collaborations, especially if we can co-develop proposals or apply for funding together. Some calls, like those from DST or DBT, allow international partners too. Let's stay in touch—I'd be happy to share opportunities as they come.

**Speaker 1:** That's great. One thing that caught my eye—your finding on a 791% increase in cis-jasmonate after hypergravity. That's remarkable!

**Speaker 2:** Yes! Cis-jasmonate is a stress hormone. When we apply hypergravity, it mimics abiotic stress and triggers higher levels of such hormones. This leads to better root systems, more auxin, and overall stronger plants. The signal response is real and measurable.

**Speaker 1:** Very interesting. Currently, our machine restricts plant height. We're thinking of building a taller version—maybe 4 meters—to handle larger crops.

**Speaker 2:** That's a good idea. You'll need to maintain balance and motion precision, but if you can pull it off, it'll really expand the crops you can test.

**Speaker 1:** One more thought—if our machine can reliably prime field crop seeds like wheat, rice, or pulses at scale, that could be a game-changer, right?

**Speaker 2:** Absolutely. If you can show quantifiable benefits—like better germination, drought resistance, or yield—then yes, it's an amazing revenue model. Seed companies are always looking for technologies that give them a competitive edge.

**Speaker 1:** That's very reassuring. By the way, have you looked into hypergravity effects on Dubia roaches or other insects?

**Speaker 2:** Not yet, but that's an interesting angle. Most research so far is on plants and small mammals, but insects could reveal some fascinating adaptations too.

**Interview with Abhijeet Dubey- expert. Owner- PS-Engineering, Bachelor of Business Administration and management.**

1. What key factors should a startup consider when choosing a revenue model to ensure long-term financial sustainability?

"That's a great starting question. I think the biggest mistake startups make is choosing a revenue model because it looks good on paper, not because it fits their business. The first thing I always look at is customer behavior—how people want to pay and how often. Then comes scalability—can the model grow with the company without needing to constantly rebuild it? And finally, I'd say cash flow timing. You need to know when money is coming in and whether it aligns with your costs. A revenue model is not just about how you earn—it's how you survive financially."

2. How do you evaluate whether a business should pursue recurring revenue, transactional sales, or a hybrid model?

"Recurring revenue is great for stability, but it doesn't suit every product or customer. Transactional sales can work well if the product has a long lifecycle or high one-time value. A hybrid model is often the most realistic—like offering a core product as a transaction and layering services on top with a subscription. I always ask: How often does your customer need you? That usually answers the question."

3. What are the financial trade-offs between targeting high-margin, low-volume markets versus low-margin, high-volume ones?

"It's really about resources and strategy. High-margin, low-volume markets give you more breathing room—you can afford mistakes. But the downside is that sales cycles are longer and customer acquisition is harder. Low-margin, high-volume models need tight operations, and they leave less room for error, but they can scale faster. I usually recommend that early-stage startups avoid the low-margin route unless they have a rock-solid cost structure and supply chain."

4. How should a startup forecast and manage its cost structure during early growth and scaling?

"Startups often underestimate fixed costs—like salaries and infrastructure—and overestimate how quickly revenue will grow. So I always suggest building two forecasts: a best-case and a conservative one. Then manage to the conservative one. Also, variable costs should scale with sales, but keep an eye on hidden costs like customer service or after-sales support that often grow faster than expected."

5. What financial metrics or benchmarks are most useful for tracking the health of an early-stage company?

"Cash burn is number one. If you don't know your runway, you're flying blind. Then I'd look at gross margin—because if that's not healthy early on, it's hard to fix later. Finally, something like customer acquisition cost versus customer lifetime value, even if it's just an estimate. You need to know if you're spending more to acquire customers than you're making back."

6. What do investors typically look for when assessing a startup's revenue potential and financial strategy?

"Investors want to see two things: traction and logic. Traction means there's some proof that the market wants what you're offering—whether that's early sales, pilots, or letters of intent. Logic is about whether your financial strategy makes sense—do you understand your market, your cost base, and your revenue streams? A flashy idea without a grounded financial plan is a red flag for most investors."

7. How should a startup decide between equity funding, debt, or non-dilutive funding like grants and subsidies?

"Each has its place. Equity is good when you're pre-revenue and need room to experiment. Debt is useful when you have predictable revenue and need cash for growth. Grants and subsidies are gold—if you qualify, go for them first because they don't cost you ownership. Ideally, a smart mix works best: use grants to build, equity to grow, and debt to scale."

8. What principles should guide a startup in setting prices for a new or innovative product or service?

"Price isn't just about cost—it's about value. I always say, start with what your customer is willing to pay, not what it costs you to make. Then work backwards. Also, don't forget to test pricing early. You'd be surprised how often companies leave money on the table just because they're afraid to ask for more. Price tells a story—about quality, trust, and usefulness."

9. How can financial modeling support strategic decisions like whether to outsource or develop capabilities in-house?

"Financial modeling shows you the real cost of a decision, not just the obvious one. Let's say outsourcing looks cheaper—but the model might reveal hidden costs in quality control or shipping delays. On the flip side, building in-house might seem expensive until you realize the long-term cost savings and IP control. A good model forces you to quantify trade-offs, not just guess."

10. What indicators should guide a startup in deciding when to scale, pivot, or seek exit opportunities?

"The biggest one is product-market fit—are customers buying, and are they coming back? If not, maybe it's time to pivot. If yes, and growth is constrained only by resources, then it's time to scale. As for exits—look for signals like industry consolidation, inbound interest, or market saturation. But the most important thing? Don't scale or exit because others say it's time. Your numbers should tell you when."

Expert on seed business

**1. From your experience, how important is innovation for staying competitive in the seed industry today?**

"Innovation is absolutely essential. The seed industry has become far more data-driven and competitive than it was even ten years ago. Farmers today expect higher yields, better resilience, and faster germination—and they want it all with fewer inputs. If a company isn't investing in innovation, it's going to fall behind very quickly. It's no longer just about selling seeds; it's about offering solutions."

**2. What types of innovations—whether in genetics, treatment, or technology—do you think are shaping the future of the seed business?**

"I'd say the biggest shifts are happening in two areas: precision breeding and seed treatment technologies. Genetic advancements—like CRISPR and other editing tools—are enabling us to develop very targeted traits. At the same time, treatments like priming and biological coatings are opening up new ways to improve crop performance without relying on chemicals. There's also a growing interest in integrating digital tools—like sensors and AI—for monitoring germination and crop development."

**3. In your opinion, what are the main barriers for startups when trying to introduce innovative solutions into the seed market?**

"One major barrier is trust. The seed industry is quite traditional, and established players—and customers—are cautious. It takes time to prove that a new product works consistently under different

conditions. Regulatory approval can also be a bottleneck, especially in Europe. And, of course, funding. Innovation is expensive. You need lab space, testing capacity, and time. Not every startup has the resources or patience to see it through."

**4. How valuable is it for startups to form partnerships with established seed companies or research institutions early on?**

"Extremely valuable. Established companies have distribution networks, credibility, and access to farmers. Partnering with them can significantly shorten a startup's learning curve. Similarly, research institutions can help validate technologies with independent data, which boosts your reputation in the market. You don't want to do it all alone—especially not in this industry."

**5. What kind of support—technical, logistical, or market access—can established players realistically provide to startups, if a good relationship is built?**

"It depends on the partner, but in most cases, it's a mix. Some might offer field trials or allow you to run pilot tests using their infrastructure. Others might connect you with their client base, or even provide joint branding. If you're really aligned, you might get access to internal R&D resources or even financial investment. The key is mutual benefit—you need to bring something of value to the table too."

**6. Do you believe that collaboration is more effective than competition in the seed sector, especially for young companies with new ideas?**

"Definitely. For startups, collaboration isn't just helpful—it's almost necessary. Competing head-to-head with legacy companies when you're small and unknown can be exhausting and risky. Strategic collaboration gives you access to scale, data, and credibility. And for larger companies, partnering with a startup brings fresh thinking. It's a win-win if done right."

**7. How critical are public or private grants for startups in the first few years of entering the seed industry?**

"They're absolutely critical. Grants give startups the breathing room to focus on R&D without immediately worrying about revenue. They also send a signal to investors that your work has been vetted. Some of the most promising technologies I've seen wouldn't have survived without early-stage funding from EU programs or innovation hubs."

**8. What role do you see governments and grant agencies playing in supporting seed innovation and scaling up new technologies?**

"Governments and grant agencies have a big role, especially in a sector like this where private investors are still a bit cautious. Public funding can de-risk innovation. It also encourages collaboration across academia, startups, and industry. The best programs don't just hand out money—they help with mentoring, networking, and validation."

**9. Finally, what advice would you give to a young company that is trying to bring a disruptive technology into a traditionally conservative industry like seed production?**

"Start small, validate often, and build relationships. Don't expect instant acceptance. Focus on one or two key use cases where your technology can really shine. Get solid data, be transparent about what works and what doesn't, and find partners who believe in your mission. Disruption is possible—but it requires patience, credibility, and a clear value proposition."

**Expert on Business Management & Startup Strategy**

**1. From your experience, how important is the choice of revenue model for long-term profitability?**

"The revenue model isn't just a financial choice—it's a strategic one. It shapes how customers interact

with you, how value flows, and how cash is generated. A poor fit between the business model and revenue structure can kill a good product. The best companies align their revenue model with customer expectations and their own cost structure."

## **2. What makes hybrid revenue models attractive for early-stage companies?**

"Hybrid models offer flexibility. Instead of choosing between a one-time sale or a subscription, you can blend both—say, charge a base fee for access and add performance-based bonuses or usage tiers. It's especially useful when your customers vary in size or buying behavior. It also helps in testing what sticks before scaling."

## **3. Are there downsides or challenges to hybrid models?**

"Absolutely. They're more complex to manage—financially and operationally. You need good systems to track usage, renewals, and performance metrics. And you need to communicate clearly with customers. Confusion leads to churn. But done right, hybrid models create predictable revenue while still capturing upside."

## **4. What key metrics should startups track when using a hybrid model?**

"There are a few critical ones: Customer Acquisition Cost (CAC), Lifetime Value (LTV), churn rate, and the ratio of recurring to variable income. It's important to balance stable income with scalability. If your variable component grows faster than your cost base, that's a good sign. You also want to monitor how revenue per customer changes over time."

## **5. Many startups struggle with early profitability. How should they balance growth and financial health?**

"You don't have to be profitable from day one, but you do need a clear path to it. Too many startups burn cash chasing top-line growth without understanding unit economics. You should know—per customer—when you break even. Profitability doesn't mean being boring; it means being sustainable."

## **6. Can hybrid models help attract investors, or do they scare them off?**

"It depends on the execution. Investors like predictability, and hybrid models—if they include a recurring component—can offer that. But you must show that you can scale it, handle the complexity, and retain customers. A well-documented hybrid model with solid traction can actually be a huge asset in fundraising."

## **7. What advice would you give a startup founder designing their first revenue model?**

"Don't overcomplicate it. Start with how your customer prefers to pay—then layer in features that align with your value. Maybe a low entry price with usage-based upgrades. Or a base subscription with optional services. Test fast, collect feedback, and adjust. A model that works in theory but confuses your buyer is no good."

## **8. Do partnerships play a role in making hybrid models profitable?**

"Big time. If you can bundle services with a partner or integrate into their ecosystem, it strengthens your value proposition. It can also reduce acquisition costs. Strategic partnerships can open new revenue channels—like referrals, white-labeled versions, or co-branded subscriptions."

## **9. What are common mistakes startups make with pricing in hybrid setups?**

"Trying to do too much at once—too many tiers, unclear value, or overpromising on deliverables. Keep it simple. Anchor pricing around results. If you're performance-based, define success clearly. Don't confuse the buyer with a menu of 12 options. Make it intuitive and aligned with outcomes."

## **10. Final thoughts for founders choosing a model in a competitive or emerging industry?**

"Focus on clarity, flexibility, and value alignment. The best revenue model is one your customer understands and is happy to pay for. Don't lock yourself into a structure just because it's trendy. The model should evolve as you grow, but the foundation should always be aligned with what the customer values most."



